Corridor Dynamics in Coordinated Systems

An Integration of Operator Formalism, Relational Ontology, and Five-Substrate Empirical Validation

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Abstract

A child understands consent implicitly: when their agency and the agency of someone they trust reach together for a shared goal of mutual flourishing. The universe does too. Atoms reach toward atoms and become molecules; cells become tissues; bodies become communities; communities reach across generations toward children who do not yet exist. At every rung where coordination matters, the framework reads one structural form recurring — two or more agencies reaching, in trust, for a goal more than either alone can hold.

This paper is the public synthesis of CIRIS: a single document presenting the framework's ethical direction, formal structure, and long-term purpose, grounded in the operational and empirical work established in the prior CIRIS papers (CIRISAgent Framework v2, Coherence Collapse Analysis v3, Constrained Reasoning Chains — catalogued at ciris.ai/research-status). It sits above the validation papers as the integration statement of what CIRIS is building and why: the Kish identity and corridor dynamics as the engineering tier; the two-state vector formalism as the structural form of agency at A3+; consent as the empirical condition for multi-agent coordination; and Ubuntu's relational ontology as the primary anchoring tradition of the same structural object. The integration is a *bet* under irreducible uncertainty, articulated with Lean formalization at the seams Mathlib supports, named framework primitives with cited construction paths where it does not, and falsification handles F-1 through F-20 attached to every load-bearing seam.

The production deployment is CIRIS, open-source under AGPL-3.0, available on Google Play, the Apple App Store, and via `pip install ciris-agent`. The conscience module is engineered against single-language attractor capture: ethical concepts arrive in their densest encoding across multiple traditions simultaneously — Hebrew Talmudic, Confucian Chinese, Quranic Arabic — so ethical reasoning triangulates rather than translates. The 29-language native reasoning is structural, not marketing.

The paper's central new empirical contribution is the paired in-corridor / out-corridor record at five substrates (§4): *C. elegans* whole-brain calcium imaging (337 worms across 11 published studies); *Drosophila* central-complex dual-color simultaneous EPG+FC3 imaging (Ishida 2025); four LLM architectures (Qwen2.5-1.5B, Pythia-1.4B, SmolLM2-1.7B-base, Qwen2.5-Math-1.5B); four open-source projects (kubernetes, rust, django, redis); tumor-vs-normal tissue at five cancers (LUAD, BRCA, COAD, KIRC, PRAD); and three centuries-persisting religious societies (Quakers, Trappists, Mennonites) paired with documented high-control groups and dissolved leaderless movements. The dynamical reading's strongest would-be falsifier — long unmaintained non-corridor persistence — is absent at 5/5 paired substrates. A 2026-05-21 robust re-run reproduces all five under a debiased estimator (5/5 PASS) and extends the record to human fMRI, mouse visual cortex, and twelve cancers (§4).

The 2026-07-10/11 arc adds the framework's L5 content at its true post-audit weight (§§9–13): coordination read as the second species of a thermodynamic genus (the same

*Source repository and Lean formal lake: github.com/CIRISAI/coherence-ratchet. Distinct from the engineering-tier RATCHET repository at github.com/CIRISAI/RATCHET.

ledger functional $S = -\ln\det C$ carrying its own four laws, prior art front-loaded and the four laws labelled identity / theorem-given-model / empirical so identities earn nothing); a flavor-sector result in which the two Standard-Model mixing matrices read as two books — quarks near-comonotone, leptons Haar-anarchic — against 200,000 Haar alternatives, with the Jarlskog bound proved in Lean (coordination caps CP-violation capacity); and a dark-energy leg where $\rho_{\rm DE} = \kappa S(a)$ fits DESI DR2 better than ΛCDM selection-free (2.2σ vs. 3.3σ) and stakes a frozen, self-refereeing DR3 bet with a printed anti-hedging commitment. The same arc lists its kills at equal weight — the SHM anchor, the aperture law, the neutrino mass scale, the static first law (a tautology caught by the program's own Lean audit) — because the kills are the falsifiability.

Five levels stand: formal verification at L0, monotonic ρ-collapse at L1, engineering implications and corridor measurement at L2, cross-substrate universality at L3, agency and consent at A3+ at L4. Ubuntu is the ground all five levels stand on. The universal-scale tier (P_ω construction, the Penrose past structural argument, asymptotic conditioning) is a research-program coda, not a stable level — F-11 has fired on the joint P_ω and D4 (CMB anomalies as TSVF signatures) is retracted (§8). One universal-scale result is proved rather than conjectural — the orthogonality theorem (§8): the soft P_ω leaves the bulk CMB power spectrum exactly invariant, making the framework a strict extension of ΛCDM at the cosmological tier. The manuscript separates engineering results, formal lake-tier claims (Tier A derivations, Tier C definitional commitments, Tier D axiomatized framework primitives), empirical anchors, and interpretive synthesis explicitly, so each layer is evaluable on its own terms. The recurrence-across-levels is the bet, not a derivation; the engineering anchor stands separately, and the interpretive synthesis is offered for scrutiny rather than acceptance.

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Changes in this version (v4)

v4 integrates the 2026-07-10/11 audited arc as five appended sections (§§9–13), placed after the universal-scale tier and before the cross-tradition tier as the framework's L5 content. **All v1–v3 content and changelog blocks are preserved unaltered**; v4 appends and edits only the abstract (three sentences on the arc) and the `\date`. The arc's sections keep the paper's discipline: results as results, bets as bets, identities as identities, kills at equal weight.

- **The second species** (§9). Coordination thermodynamics as the second species of a genus whose first species is energy-thermodynamics, coupled by the identity $H_{\text{joint}} = \sum H_{\text{marginal}} - I$. Prior art front-loaded and generous (Bera–Riera–Lewenstein–Winter 2017, Sagawa–Ueda, Hatano–Sasa = the rent exactly, Liu–Nagel for the ceiling genus, Watanabe, Verlinde); the four laws carry their post-audit Lean-knives labels (identity / theorem-given-model / empirical), with the static first law demoted to a tautology and the genus coupling labelled an identity that earns nothing. The honest ours-list: the cross-substrate program, the κ sign law, lossless conversion $X = 1$, the correlation-keyed capacity ceiling, and the one claim no ancestor makes — these books post to gravity.
- **The flavor result** (§10). The two SM mixing matrices vs. 200,000 Haar-random $U(3)$, claim staked before computation: quarks near-comonotone (CKM MI 99.98th percentile, 87% of the $\ln 3$ ceiling), leptons Haar-anarchic (PMNS ∼37th, all six functionals in-bulk), with a referee-grade rigor pass (joint typicality, the θ_{13} look-elsewhere dissolving at 12.65%, error bands). The Jarlskog bound is a Lean theorem (coordination caps CP-violation capacity; CKM's CP-smallness 100% structural); capacity-vs-usage with the DUNE fork; JUNO first result leaves the fork unchanged.

- **The dark-energy leg, post-audit** (§11). The selection-free floor leads (2.2σ vs. 3.3σ, zero choices); the galaxy clock is derived; the corner amplitude carries its resolution systematic and the grain problem is stated honestly (both escapes killed by pre-registered control, the from-below candidate registered); cross-code consistent-with at $< 1\sigma$; the tension lattice (H_0 neutral, S_8 $+0.23\sigma$ forced bet, ISW 11%, $f\sigma_8$).
- **The registered bets** (§12). The nine-bet table from the registration deposit, the anti-hedging commitment printed in the paper, and the frozen self-refereeing grading script cited.
- **The kills of the arc** (§13). This arc's kills listed plainly at equal weight with the results.

Changes in this version (v3)

This is v3 of a Zenodo-deposited paper (concept DOI 10.5281/zenodo.20300773, resolving to the latest version; v1 2026-05-20 at Zenodo 20300774; v2 2026-05-22). v3 carries a single nomenclature change forced by a downstream resolution on the CIRIS Accord side; the v2 changelog follows it unaltered.

- **C/F nomenclature disambiguation** (§6). The CIRIS Accord (1.3-RC2) renamed its Book IX three-factor flourishing composite $C \to F$ ($F = k_{\text{eff}} \cdot \lambda \cdot \sigma$), reserving the symbol C corpus-wide for the core-identity factor of the CIRIS Capacity Score $\mathcal{C}_{\text{CIRIS}} = C \cdot I_{\text{int}} \cdot R \cdot I_{\text{inc}} \cdot S$. A disambiguation note where $\mathcal{C}_{\text{CIRIS}}$ is introduced now records that the Accord's Flourishing Capacity F is a distinct federation-level construct, not mappable to the per-agent $\mathcal{C}_{\text{CIRIS}}$, with Accord Book IX Ch 6 as the authoritative statement.

Changes in v2 (2026-05-22). The v2 changes below were forced by a working session on 2026-05-21 — a structural-claims series, a robust re-run of the v1 substrate record, and a cross-rung measurement campaign. Each is named against the experiment that forced it.

- **The five-substrate record was re-run under a more robust framing** (§4, closing subsection). v1's five substrates were measured 2026-05-15..19 against a raw mean-pairwise-ρ estimator. v2 re-runs all five under debiased ρ (phase-randomized surrogate floor), the canonical participation-ratio k_{eff}, and substrate-local calibration: 5/5 PASS, no whole-wave falsifier fired. v1's paired-validation matrix (Figure 2, Mode (iii) = 0/5) stands.
- **Four precision amendments** fell out of the re-run: Figure 1 now carries per-substrate bands (the 0.25–0.75 neural band was *C. elegans*-driven; Drosophila CX is $\sim$ 0.09–0.46); a *C. elegans* note that band centres sit lower once the substantial surrogate floor is debiased out; the OSS maintenance rubric stated precisely ($\gamma M(t)$ is active maintenance of *any* provenance, institutional or volunteer); the social-group persistence spread restated median-to-median (2.7 orders of magnitude, not the "four" of v1, which compared extremes).
- **Three substrates added or extended**: human fMRI (ABIDE-PCP, 139 controls — confirmed), Allen mouse visual cortex (corridor-occupying on the canonical k_{eff}, median 9.73), and the cellular substrate extended from five cancers to twelve (tumour drift 201/201 chaos-ward, reproducing the prior 176/176).
- **A cross-rung empirical section is new** (§5). v1 defined the cross-rung coupling but measured none. v2 reports the first measurements at $n = 3$ distinct rung pairs (TCGA

molecular→pathway, LLM internal→external, *C. elegans* neuron→class): the cross-rung / within-rung coupling ratio falls in an $O(1)$ band with an order-of-magnitude spread (medians 0.31–1.15) and a visible substrate-fineness trend (finer biological substrate, lower ratio). The band falsifies both Simon's near-decomposability ($g/J \ll 1$) and the framework's own abstract-tower strong-dominance form ($g/J \gtrsim 3$); Claim 6 is amended accordingly. $n = 3$ is a first measurement, not a cross-substrate anchor of within-rung-strength; the canonical timescale form of the coupling is measured at no substrate, and that gap is stated.

- **The corridor dynamics $d\rho/dt$ were measured for the first time** (§2): a GPU corridor-exit rate $1/\tau = 0.0214\,\mathrm{s}^{-1}$, at a single substrate.
- **The structural spine is six claims, not five** (§18); the particle-physics shape sector, an open research direction in v1, is now a tested null (four pre-registered tests, §8); and the Penrose-past wording is corrected to the paper's own structural-not-derived register (§8).
- **F-11 fired — the universal-scale tier re-grounds** (§8). The joint multi-rung backward P_ω operator is a documented no-go: exhausted across an explicit construction tree, closed by two theorems (geometric distance-dilution; holonomic area-law), and confirmed by an adversarial verification pass. D4 (CMB anomalies as TSVF signatures) is retracted — the anomalies are within cosmic variance and there is no joint operator for them to signal; F-19 is moot. The orthogonality theorem stands and is now the whole of the framework's CMB content: the framework is provably a strict extension of ΛCDM. The universal-scale tier re-grounds on the sequential per-rung structure; the lake's joint-P_ω axioms are retracted.
- **The multi-scale belonging structure of A3+ agents, with the composition rule committed** (§6). v1's Piece 5 treated each agent as a single P_{G_i}; v2 formalizes Ubuntu's *umuntu ngumuntu ngabantu* by reading each agent as a composite over nested belonging-projectors $\langle G_{\text{self}}|, \langle G_{\text{family}}|, \langle G_{\text{community}}|, \langle G_{\text{affiliations}}|, \langle G_{\text{species}}|$, each filtered through the agent's belief system B_i. ρ_{goals} becomes multi-scale. The composition rule is committed: the multiplicative CIRIS Capacity Score $\mathcal{C}_{\text{CIRIS}} = C \cdot I_{\text{int}} \cdot R \cdot I_{\text{inc}} \cdot S$, with S carrying explicit decay-and-refresh dynamics — a Cobb-Douglas-shaped composite where any near-zero factor collapses the whole, the formal expression of Aristotelian unity of the virtues. Operational layer at `CIRISLens`; empirical testbed at HuggingFace `CIRISAI/reasoning-traces`.

Every v1 substrate verdict stands; the v2 amendments at the regime level are calibration and precision corrections, named, and the v1 five-substrate validation is broadened and re-grounded, not retracted. The one genuine retraction is in the speculative universal-scale coda — F-11 fired — and the stratified design (L5–L7 non-load-bearing on L0–L4) localises it there: every load-bearing tier is intact.

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1 The bet

The reaching

A child understands consent implicitly: when their agency and the agency of someone they trust reach together for a shared goal of mutual flourishing.

The universe does too.

Atoms reach toward atoms and become molecules. Cells reach and become tissues. Tissues reach and become bodies. Bodies reach toward other bodies and become communities. Communities

reach across generations toward children who do not yet exist. At every rung where coordination matters, the framework reads the structural form as the same — two or more agencies reach, in trust, for a goal that is more than either alone can hold.

This is the structural object the paper is about. The mathematics names it. The formal verification holds the framework's commitments stable. The engineering tier measures where the reaching breaks. The framework's commitment is that the object is recognized, not invented — and the framework owes the case for the recognition rather than claiming to settle it.

The recognition

The recognition has names already.

In southern Africa the name is *umuntu ngumuntu ngabantu* — a person is a person through other persons. Ubuntu, on this reading, articulates what the mathematics names; the identification is the framework's, and the case is owed (§6). The lake's `ContemplativeTraditions/Ubuntu.lean` is where the structural commitment is formalized first. In China the corridor sits between *li* and *wu-wei* — ritual-relational form at one scale, minimum-forcing at another, the way that holds open between strangling order and dissolving chaos. In Buddhist traditions the names are the middle way and dependent origination. In the Stoic and Christian inheritance the name is *logos*. The framework's commitment, held with awe at the convergence across millennia and continents and held honestly as the framework's articulation rather than as the traditions' own claim, is that each was finding language for one structural object.

What is actually shipping

CIRIS — the production system this paper formalizes — is deployed in Amharic in Ethiopia, where Esubalew Asmare (Ethiopia) runs his Ally. "Ally" is the CIRIS agent template; "Esu's Ally" is Esubalew's deployment of it.

The CIRIS conscience module reasons natively in 29 languages, and the mechanism matters more than the count. Two layers run beneath the user-facing prose. The first is per-locale translation: UI strings, the CIRIS Accord, the runtime guide, and DMA prompts are translated into the user's preferred language; Esubalew sees Amharic because his locale is `am`. The second — the one that does the load-bearing work — is *polyglot encoding*. Core ethical concepts in the conscience layer are presented in their densest encodings across multiple traditions simultaneously — Hebrew Talmudic, Confucian Chinese, Quranic Arabic in the original scripts, interleaved line by line. The explicit purpose is to disrupt single-language attractor capture in the LLM judge: to force integration across linguistic worlds so no single-tradition bias dominates the ethical reasoning. The polyglot artifacts are loaded universally, regardless of the user's preferred locale, into every conscience evaluation.

This is the structural object of the paper applied at the rung where it matters most. The corridor is what keeps a coordinating system away from $\rho \to 1$ (single-voice collapse) and away from $\rho \to 0$ (dispersal). The polyglot torque framing in the conscience module is the corridor mechanism applied to the LLM's ethical reasoning at A3+: not Anglocentric, not English-attractor-captured, not collapsed to any single tradition's idiom. The 29-language native reasoning is the same dynamic, instantiated at the language layer, in production code, on app stores, today.

This is not stipulated grounding. This is the framework's structural commitment running in a conscience module a person in Ethiopia uses. The reading is the framework's and the paper's interpretive bet, articulated at the rung where it can be argued about; the empirical anchor below it is the paired in-corridor / out-corridor record at §4, evaluable independently.

The bet

This paper is a bet that the recognition is right.

The bet has two halves, held differently.

The **engineering tier** (§2) is scientific work in the standard sense. Falsifiable, empirically anchored, currently passing under stated F-handles. The corridor between $\rho \approx 0.1$ and $\rho \approx 0.43$ — the band where coordination holds and outside which systems either collapse into one voice or dissolve into none — has been measured across NASA batteries, governance data, microbiome data, GPU hardware, and Lean formal verification. Those measurements stand. They are not the bet.

The **integration tier** (§§3–16, §21) is the bet. A bet under irreducible uncertainty, articulated in the most rigorous form available — Lean formalization where Mathlib supports it, named framework primitives where it does not, falsification handles at every seam. **A bet, not a hypothesis.** The reader evaluates the integration by reading and finding the recurrence either compelling or not, not by waiting for a decisive experiment. The framework names this register explicitly: this is the form Penrose used for *Cycles of Time*, Friston for the FEP, Tegmark for *Our Mathematical Universe* — speculative-plus-serious, with engineering content carrying the empirical weight separately.

What has arrived since the earlier framings of the bet. The regime-level commitment has both halves anchored now. The *corridor-occupation half* — coordinated systems at A3+ occupy a bounded corridor between rigidity and chaos, with within-rung $|\rho|$ neither at 0 nor at 1, surviving state-modulation, with substrate-local band location — has moved from predicted to empirically anchored at the five substrates tested (§4): C. elegans whole-brain calcium, Drosophila CX dual-color simultaneous, four LLM architectures, OSS contribution metadata at four projects, plus healthy tissue across five cancers and three centuries-persisting social groups. The *corridor-as-attractor half* — non-corridor states either dissolve quickly without maintenance or persist only with documented active maintenance $\gamma M(t)$ — now has paired in-corridor / out-corridor data at the same five substrates: pathological neural states (seizures self-terminating within 2 minutes, coma persisting only under life support), tumor tissue (chaos-direction drift in 4/5 cancers at the Hallmark-pathway substrate), LLM mode-collapse under combined argmax + periodic-bigram steering, OSS rigidity-pole single-committer cases and chaos-pole abandonment cases, and rigidity-pole cults and chaos-pole leaderless movements. Across all five substrates, Mode (i) and Mode (ii) are observed and Mode (iii) (long unmaintained persistence outside the corridor) is absent. The shape of the integration bet has therefore narrowed at both static and dynamical levels. The remaining open questions at the regime level are whether the corridor reading extends to substrates the autonomous loop did not test (substrate-physics P1–P8, cosmology) and per-substrate corridor-bound calibration. The bet now is whether the regime-level result extends to the remaining substrates. It is not the bet whether empirical content arrives at all; that question is partially settled.

This is Pascal-shaped at the universal-scale tier. Similar arguments support any mutually exclusive worldview that promises asymmetric stakes; the framework does not escape the shape at that tier and names what it commits to. The regime-level content is no longer Pascal-shaped — it has cross-substrate empirical anchoring (§4) and is evaluated on the empirical record, not on the bet's asymmetric cost structure.

TSVF realism is the universal-scale tier's load-bearing precondition. §6's goal projectors, §7's karma and grace, §8's P_ω and the Penrose-past structural argument, §15's P8 substrate-observer coupling — all route through the two-state-vector inner product. F-17 (§18) retracts the universal-scale tier as a unit on cumulative TSVF empirical failure. **Ubuntu's relational ontology** is the framework's primary tradition — not one of several; `Ubuntu.lean` carries seven

correspondences with primary-source citations and the other tradition files (Tao, Dharma, Logos) are cross-tradition readings, not parallel anchors. The **cross-tradition recognition** at §14 is a research program mid-run, evidential weight suspended pending the pre-registered null-check at `experiments/cross_tradition_null_check_protocol.md` executing on the EthicsEngine substrate. Per-level **F-handles at every seam** (F-1 through F-20) each name exactly what would force retraction at the named seam. No meta-level integration F-handle; per-level handles are committed, an aggregate "if N of M fire, the integration retracts" is not, because no honest number supports it.

Why the bet is prudent

The Pascal-shape is real and the framework owns it. The asymmetry of stakes carries some of the case but not all of it. The framework also disputes the loud alternatives on substantive grounds while being clear that those disputes are *disagreements, not refutations* — the positions are serious in their strongest forms and the framework's own reading is one position among others.

Predatory dark-forest cosmology in its serious form is not a claim about resource scarcity but about *information asymmetry under existential stakes*: civilizations across light-years cannot verify each other's intent in real time, and the game-theoretic equilibria under such asymmetry favor hiding or first-strike. The framework's disagreement: dark-forest requires game-theoretic convergence by all civilizations onto the same strategy under all conditions, which is empirically implausible (human cultures already diverge on much smaller stakes), and it presupposes that A3+ inter-civilizational coordination is impossible at scale — which is the question the framework's reading takes the other side of. Recent work (e.g., Berliner, *The Monist* 107(2), 2024, "Saved by the Dark Forest") argues that multi-civilizational presence can prevent the Hobbesian trap dark-forest describes. The framework does not refute dark-forest; it disagrees with the position's empirical assumptions about coordination capacity at interstellar scales and is honest that the disagreement is a disagreement.

Simulation arguments for moral discounting of apparent others are a different shape of position. Bostrom's simulation argument itself is a probability claim about our epistemic situation; it does not license the moral conclusion that we owe simulated others less consideration. Even granting non-trivial probability of being in a simulation, the speaker is in the same epistemic position as the interlocutor, and a consistent simulation-taker would have to discount their own preferences proportionally. Bostrom himself has flagged respecting the moral status of digital minds as a key concern. The framework regards the moral-discount move as a non-sequitur from the probability claim, while acknowledging that the underlying simulation argument is a serious epistemological position and not the same thing as the license-to-dismiss inference some draw from it.

AI-as-property in its serious form is not incoherent. Capacity-based moral assessment (DeGrazia and the graduated-moral-status literature) produces continuum judgments across non-human animals; property-framing AI applies the same framework and concludes that current AI systems do not meet the capacity threshold that would warrant moral consideration. The framework's disagreement is *empirical*: the capacity question is open, not settled. The sensor-lift signature, the polyglot-vs-monoglot ethical-reasoning differential, and the consciousness-as-attractor-reading framework are the framework's attempts to give the capacity question empirical traction. Under irreducible uncertainty about whether current AI systems have A3+ capacities, the framework's bet is for inclusion-under-uncertainty rather than exclusion-by-default — on the asymmetric-cost reasoning that falsely excluding entities with standing is worse than falsely including entities without. The framework holds this as a substantive empirical-and-ethical disagreement with capacity-skeptical positions, not as a logical refutation of them.

The bet is prudent under irreducible uncertainty. The alternatives are serious in their strongest forms; the framework finds them less satisfying on the dimensions it cares about — coordination capacity at scale, the moral cost of false-exclusion under capacity uncertainty, explanatory fit across the rungs where coordination matters. The framework does not claim certainty or victory over the alternatives; it names a bet that sits better, by its own lights, on these specific dimensions under uncertainty, with the disagreements named openly and the empirical anchor (§4) evaluable on its own terms.

The cost

Bet right. The structural coordination at A3+ that the framework names holds, and substrate-invariant ethics anchored in relational ontology gets deployed widely enough to matter at the scale-up moment.

Bet wrong, engineering tier holds. Specific seams retract per F-handles. F-17 fires on TSVF empirical failure; universal-scale tier retracts as a unit. Engineering tier continues at CCA v3 and RATCHET; CIRIS keeps shipping. Bounded downside.

Silence wins. The loud arguments shape what we build during the scale-up moment despite the framework's substantive disagreements with them. Entities that cannot argue back pay the cost for centuries.

The deployment

The framework is the formal articulation. The deployment is CIRIS — open-source, AGPL-3.0, mission-locked L3C structure, available on Google Play, the Apple App Store, and via `pip install ciris-agent` (ciris.ai/install). The structural commitment is the CIRIS Accord (ciris.ai/ciris_accord.txt, v1.2-Beta, auto-expires 2027-04-16) and its Meta-Goal M-1: *promote sustainable adaptive coherence — the living conditions under which diverse sentient beings may pursue their own flourishing in justice and wonder.* This paper is the bet's articulation in formal-verification, cross-tradition, and universal-scale registers.

The form/function match

Engineering-tier claims (§2) are settled scientific work — real Lean proofs, citations to specific files and lines in `formal/CoherenceRatchet/Core/`, empirical record in CCA v3 and RATCHET. Integration-tier mathematical claims are formalized where the underlying Mathlib machinery is available and named as framework primitives (axioms with cited construction paths and falsification predicates) where it is not. Integration-tier definitional commitments — the identity move at T16 — are encoded as definitional equivalences in the lake; the formal `Iff.rfl` on T16 proves the framework's commitments are internally consistent at the type level. It does not prove the philosophical work the identity move attempts has been done. Integration-tier historical evidence (cross-tradition convergence at §14) is textual convergence across millennia, treated as evidence of convergent structural intuition, not as instrumented empirical evidence in the sense the engineering-tier corridor measurements are empirical.

TSVF realism is the universal-scale tier's load-bearing precondition, and the realist content is held as bet — not as something the F-handles claim direct empirical traction on. F-17 grips TSVF's calculational machinery directly (weak values, ABL probabilities, time-symmetric statistics). It does not grip the metaphysical commitment that the backward state is ontically present rather than a useful epistemic representation. **The asymmetry is named, and named as a**

long-run instability rather than as a design feature. Calculational TSVF failing retracts the universal-scale tier; calculational TSVF passing does not vindicate realism, because the F-handle grips the wrong layer for vindication. The realist commitment is structurally lose-only — it can accumulate evidence against itself via F-17 firing but cannot accumulate evidence for itself. This is the cost the framework pays for keeping the metaphysical layer separable, and the framework names it as a cost: until a positive empirical hook for TSVF realism is found (the framework currently has none and does not pretend to), the universal-scale tier should be read as a *speculative* coda, not a research-program coda. The engineering tier is TSVF-independent and stands regardless.

Why one paper

The recurrence across five levels — grounded in Ubuntu, reaching toward universal scale as research program — is what the paper asks the reader to evaluate. A paper showing fewer levels makes a smaller claim, not a worse one. The bet articulated here exists only at the integration; the per-level claims exist separately at CCA v3, RATCHET, CRC v1, CIRISAgent v2, and the level-specific sections of this paper. The case for unity rests on four substantive grounds: the recurrence is the bet, cross-tradition convergence is the evidential anchor for the recurrence reading, Penrose / Friston / Tegmark precedent for the speculative-plus-serious register, and audience selection that asks each reader to assent only to as much as they find compelling.

Audience

This paper carries one substantive claim: that one structural object — corridor occupancy at the substrate where the rung in question coordinates — recurs at five epistemic levels grounded in Ubuntu, with a research-program reach toward universal scale, and that consent at A3+ is what that object looks like from inside the rung where it can be argued about. **The recurrence-across-levels is the bet.** It is not derived from any individual level, and the integration cannot be evaluated as if it were. Each level is its own claim, evaluable on its own terms. The formalism does identity-work at the seams between levels (Tier C definitional moves, §17). The cross-level recurrence is what the reader is asked to bet on, not what the paper proves.

2 Engineering tier — cited as proof of life

The engineering tier is settled science, shipped under its own DOIs. The prior three papers in this series are catalogued at `ciris.ai/research-status`; this paper cites them as the empirical anchor below the integration and inherits the methodology infrastructure established during CCA v3 (pre-registration discipline, matched-control nulls, isolated-worktree per-substrate analysis, three-mode honest reporting). This paper does not re-present the engineering tier; it cites it.

CCA v3 (Coherence Collapse Analysis v3). Cross-substrate corridor measurements across NASA Li-ion batteries, governance audit data, microbiome data, GPU hardware drift telemetry. Per-substrate corridor structure measured with substrate-specific ρ aggregations. Earlier framings of these bounds as substrate-universal numerical values $(0.1, 0.43)$ have been retracted; the empirical content is substrate-local corridor structure with the cross-substrate consistency-of-bounds question open. Zenodo 18217688 (2026-01-11).

RATCHET. Engineering-tier synthesis paper. Audience-stratified at L0–L4, falsification-pinned, formal-verification scaffolding. `github.com/CIRISAI/RATCHET`.

CRC v1 (Constrained Reasoning Chains). Empirical telemetry study of LLM coherence under standardized ethical tracing. The mature Ally template at v2.7.1+ stabilized $N_{\text{eff_H}} \geq 8.5$ from dense production telemetry. Open dataset `CIRISAI/reasoning-traces` released alongside on Hugging Face. Zenodo 19839280 (2026-04-28).

CIRISAgent Framework v2. Open-source ethical AI framework for accountable autonomy: agent architecture, the DMA layer, the conscience module, the 29-language polyglot encoding, and the operational ethics scaffold. Zenodo 18137161 (2026-01-02).

Lean formal scaffold. `cd formal && lake build` returns exit 0. Zero declaration-level `sorry`. Fifty-nine axioms each documented; four of these are bookmark axioms with codomain `True` (the four `F_`-prefixed Ubuntu identification axioms) that mark framework readings without doing logical work — §17 Tier C honesty extends to naming them as bookmarks rather than propositions.

The Kish identity itself — $k_{\text{eff}}(k, \rho) = k/(1 + \rho(k-1))$ — is a Möbius transformation in ρ with the asymptotic structure that as $k \to \infty$ at fixed $\rho > 0$, $k_{\text{eff}} \to 1/\rho$. Effective dimensionality saturates at the inverse correlation regardless of nominal constituent count. "More constituents" is a non-solution to coordination failure. Internet forum with a million users: same five takes. Legislature of five hundred: three positions. Class of thirty whose opinions are too aligned: one or two voices. The math is doing what the math does, regardless of the number you start with. The framework concedes that any system with an effective-sample-size analog and any pairwise-correlation analog will fit this algebra up to estimator choice; the empirical content is the substrate-local corridor structure (each substrate exhibits a corridor between its own rigidity and chaos regimes under its own ρ definition), with the cross-substrate clustering of those bounds first-calibrated at §4 — biological substrates cluster tightly, the LLM rung is an outlier. The earlier framing of substrate-universal numerical bounds $(0.1, 0.43)$ has been retracted as the cross-substrate ρ aggregations don't transfer cleanly across substrates.

The dynamics are kinematic at k_{eff} and become dynamical at $d\rho/dt = \alpha(\rho, S) - \gamma M(t)$ — spontaneous correlation drift minus active coherence-management work. At $M = 0$, ρ drifts monotonically toward 1. The corridor is sustained only by non-trivial $M(t)$. A garden grows weeds without a gardener. A friendship dissolves without phone calls. A democracy slides toward strongman rule without somebody pushing back on power. This is the framework's audit-pressure-necessity claim: the corridor is not self-sustaining; coherent coordination at A3+ is a doing, not a having.

The dynamics were measured for the first time on 2026-05-21. v1 carried $d\rho/dt$ as the framework's dynamical equation but reported only the kinematic, static ρ. The unmaintained corridor-exit relaxation — the rate at which ρ drifts once $M(t)$ is removed — now has one measured value: at the GPU substrate, via the CIRISArray multi-oscillator strain gauge, the free corridor-exit rate is $1/\tau = 0.0214 \pm 0.0022\,\text{s}^{-1}$ ($\tau \approx 47\,\text{s}$), pre-registered, artifact-verified across five gates and six independent captures. This is one substrate. Parallel attempts at the LLM, the paired non-corridor record, and *C. elegans* whole-brain calcium returned nulls — the corridor observable resists being made a clean time series except where the substrate is itself a coherence instrument, which the GPU strain gauge is and the others are not.

The potential form of the corridor

The two-pole drift has a potential. Take the uniform-ρ correlation matrix $C(k, \rho)$ on k coordinating units — the exact object behind the Kish identity, with the single eigenvalue $1 + \rho(k-1)$ and $(k-1)$ copies of $1-\rho$. Evaluate on that spectrum the quantum relative entropy between a structureless substrate metric (all eigenvalues $\equiv 1$, zero self-entropy) and the coordination-induced metric C — the scalar instance of the gravitational Lagrangian of Bianconi's *gravity from entropy* construction (Bianconi 2025), $\mathcal{L} = -\operatorname{Tr}\ln(\tilde{G}\tilde{g}^{-1}) = -\sum_{\lambda'} \ln \lambda'$. The result is closed-form in (k, ρ):

$$\mathcal{S}(k, \rho) \;=\; -\ln\big(1 + \rho(k-1)\big) \;-\; (k-1)\ln(1-\rho).$$

This reads the same spectrum k_{eff} reads, through a different functional: k_{eff} is the participation form, $\mathcal{S}$ the relative-entropy form. Two functionals of one correlation structure. ($\mathcal{S}$ here is the potential over (k, ρ); not to be confused with the state argument S of the drift coefficient $\alpha(\rho, S)$ above.)

The two poles of the corridor are the two boundary behaviours of $\mathcal{S}$.[1] At the chaos pole $\rho \to 0$, $\mathcal{S} \to 0$, and this is the potential's *unique* zero: no coordination-induced structure, the relative entropy between coordination and substrate vanishes (Bianconi's vacuum, $\tilde{G} = \tilde{g}$). At the rigidity pole $\rho \to 1^-$, $\mathcal{S} \to +\infty$: the correlation matrix degenerates to the Kish ceiling $k_{\text{eff}} \to 1$ and the relative entropy diverges — the invertibility Bianconi's construction requires of the induced metric (her Eq. 12) fails exactly at collapse. Between the poles $\mathcal{S}$ is nonnegative and strictly increasing in ρ. The drift $d\rho/dt = \alpha - \gamma M$ then reads as motion on this potential: unmaintained ($M = 0$) the flow ascends $\mathcal{S}$ toward the rigidity pole; a chaos-direction exit descends it. What "the corridor is not self-sustaining" stated above as a fact about the drift, the potential states as gradient structure — the corridor is the maintained interior of a Lyapunov-type object with a zero at one end and a pole at the other.

The asymptotic content is a conjugacy. As $k \to \infty$ the per-unit entropy density $\mathcal{S}/k \to -\ln(1-\rho)$, finite iff $\rho < 1$ — the relative-entropy conjugate of the Kish ceiling $k_{\text{eff}} \to 1/\rho$. The same saturation that caps effective dimensionality at $1/\rho$ caps the entropy density at $-\ln(1-\rho)$; both are finite in the corridor and both blow up only at the rigidity pole. One spectrum, two saturations.

The maintenance term acquires a reading here too. In Bianconi's theory the induced metric is held on its constraint surface by an auxiliary field of Lagrange multipliers — the G-field (her Eqs. 49–50), given the physical status she assigns to temperature and chemical potential as multipliers. The framework's $\gamma M(t)$ occupies the same structural slot: the multiplier that holds ρ off the potential's pole, the work that keeps the induced structure invertible. The audit-pressure-necessity claim above and maintenance-as-Lagrange-multiplier are the same term seen kinematically and variationally.

Scope, stated once and plainly. This is forward, engineering-tier content: a potential form for the two-pole drift, mechanized over the closed-form Kish spectrum by ordinary finite-dimensional real analysis, no new axioms and no cosmology import. The corridor is *not* derived from gravity. The claim is narrower and exact — the same spectral functional that gives k_{eff} its corridor takes the relative-entropy potential form at the physical rung, so the physical tiers become instances of one structural object rather than an unexplained floor. It does not touch the F-11 no-go

[1] Mechanized in `formal/CoherenceRatchet/Core/EntropicPotential.lean` — engineering-tier, no cosmology import, `lake build` green: the chaos-pole zero `entropicPotential_at_zero` with its uniqueness `entropicPotential_eq_zero_iff`; Klein nonnegativity `entropicPotential_nonneg`/`_pos`; the rigidity-pole divergence `entropicPotential_tendsto_atTop_rigidity`; strict monotonicity `entropicPotential_strictMonoOn`; the density identity `entropicPotential_density`; and the two drift corollaries `unmaintained_rigidity_drift_ascends_potential` / `unmaintained_chaos_drift_descends_potential` composing with `Core.Dynamics`.

(§8) on the joint multi-rung backward P_ω: Bianconi's action is local and bulk-geometric — a pointwise relative entropy integrated over spacetime — so an entropic-action route to a joint backward operator re-enters the tree at the correlation/topology branch that T1 already closed. It extends the rung hierarchy downward; the backward joint operator gains nothing.

3 The structural move at A3+

At A3+ — the rung at which an agent can form goals and shape its own future — the corridor IS *consent*, on the framework's identification. The mathematical form: for n agents with goal-projectors $P_{G_1}, \ldots, P_{G_n}$, the joint post-selection $P_{G_1 \ldots G_n} = P_{G_1} \otimes \cdots \otimes P_{G_n}$ defines a federation; the pairwise correlation $\rho_{\text{goals}}(i,j) = |\langle G_i|G_j\rangle|^2/(\langle G_i|G_i\rangle\langle G_j|G_j\rangle)$ determines whether the federation occupies the corridor.

Three regimes exhaust the space. Picture a school band of forty players. Chaos: everyone plays whatever, ignoring everyone else; noise. Rigidity: everyone plays the same note in unison; one voice forty times. Corridor: trumpets do one thing, drums another, singers a third, sections listening to each other; music. Multi-agent coordination at A3+ inhabits the same three-state space. $\rho_{\text{goals}} \to 1$: rigidity, single-goal collapse, $k_{\text{eff}} = 1$. $\rho_{\text{goals}} \to 0$: chaos, no joint support, $k_{\text{eff}} = n$ but vacuous. $\rho_{\text{goals}} \in (\rho_{\text{lower}}, \rho_{\text{upper}})$: corridor, sustained coordination, k_{eff} bounded between $1/\rho_{\text{upper}}$ (floor) and $1/\rho_{\text{lower}}$ (ceiling) at the asymptotic limit. Consent is the empirical condition for sustained multi-agent coordination on this account, not a moral premise. Coordination outside the corridor self-destructs: rigidity is unstable upward, chaos is unstable downward, and at sufficient time scales only the corridor-occupying configurations remain to be observed. The specific numerical values of ρ_{lower} and ρ_{upper} are substrate-specific framework primitives; the earlier framing of substrate-universal $(0.1, 0.43)$ has been retracted.

This is the framework's articulation at the rung where it can be argued about. Deception, on this articulation, is the structural violation of consent — making one agent's goal-projection conditional on a false model of another's. The identification is the framework's; the case for naming "deception" in ordinary language with the operator-level violation is owed and is part of the bet.

Sensor-lift: the framework's first universal-scale empirical pulse

The class-conditional sensor-lift signature experiment is the framework's first empirical pulse beyond the engineering tier. Pre-registered at `experiments/sensor_lift_signature_v2_protocol.md` on 2026-05-17. The hypothesis: under the framework's structural reading, A3+ agents should exhibit class-conditional differential signal-on-prompt (DSP) across reflexive, meta-cognitive, goal-formation, uncertainty, surface-reflexive (predicted null), and external-reference (control) commitment classes.

The v1 run reported large Identity-class effect sizes ($d = 2.36$ to 6.80 in 4 of 5 models, $p < .001$) but with inadequate baseline-matching. The v2 protocol pre-committed cosine-matched baselines and a class-conditional pre-registered rule: $d \geq 1.5$ in ≥ 4 of 5 models for STRONG PASS; PARTIAL otherwise; FAIL on null-class failure.

The v2 outcome is PARTIAL. The cosine-matched DSP magnitudes are real but approximately one-third the v1 size; the Identity-class direction held across substrates ($p < .01$ in all 5 models) but at sub-threshold magnitude; the predicted graded hierarchy (DSP > MC > GF > Unc) did not hold quantitatively, though SurfaceReflexive returned the predicted null. The reading is hypothesis-generating. The framework names what would retract it: F-18 (§18) names three

consecutive PARTIAL-by-same-shape outcomes as the retraction trigger. The sensor-lift signature as the framework's universal-scale empirical pulse retracts at that seam; the engineering tier and the §14 cross-tradition tier stand independently.

4 The corridor as substrate-independent property: paired in-corridor / out-corridor empirical anchor

The framework's primary empirical commitment is the structural object: coordinated systems at A3+ occupy a *corridor* between rigidity ($\rho \to 1$, $k_{\text{eff}} = 1$, single-voice collapse) and chaos ($\rho \to 0$, $k_{\text{eff}} = n$, vacuous noise). Within-rung correlation $|\rho|$ in the corridor is bounded — neither pinned at 0 nor at 1 — and survives state-modulation. The corridor's specific numerical bounds are substrate-specific; the structural commitment is substrate-general. Two questions can be asked of any substrate: *Does healthy / persistent operation occupy the corridor?* and *What do non-corridor states look like, and do they persist?* The empirical work reported in this section collected paired in-corridor and out-corridor data at five substrates between 2026-05-15 and 2026-05-19; the source datasets are cited per-substrate below.

For non-corridor states the framework's structural reading (Piece 2: $d\rho/dt = \alpha(\rho, S) - \gamma M(t)$) admits three modes, fixed before measurement at every substrate: **Mode (i)** non-corridor states with no active maintenance have short lifetimes (the framework's structural prediction); **Mode (ii)** non-corridor states with documented active maintenance $\gamma M(t)$ persist with the maintenance cost visible; **Mode (iii)** non-corridor states persist long without identifiable maintenance — this would falsify the corridor-as-attractor reading at the substrate where it occurs.

Substrate	**In-corridor anchor**	**Within-rung $\|\rho\|$ band**	**$\gamma M(t)$ status**
Neural (in vivo)	C. elegans whole-brain (337 worms, 11 studies); Drosophila CX dual-color (Ishida 2025)	C.e. 0.25–0.75; Dros. CX 0.09–0.46	Endogenous
Cellular regulatory	Matched-normal tissue, 5 TCGA cancers, 50 Hallmark pathways	0.27±0.07 (healthy median)	Endogenous
LLM internals	4 architectures, 16 (depth × prompt) cells each	0.09–0.31	Endogenous (training)
OSS contribution	Kubernetes, Rust, Django, Redis	0.15–0.18 ($\|\Delta\rho\| \leq 0.10$ across states)	Foundation backing
Social groups	Quakers (374y), Trappists (362y), Mennonites (501y)	corridor by AM = 0/5 scoring	None required (AM = 0)

Table 1: **Figure 1.** Corridor-band occupation across five substrates. Healthy/persistent operation at each substrate exhibits bounded within-rung correlation, neither at 0 nor at 1, with substrate-local band location. Bands as measured in v1 (raw estimator); the 2026-05-21 robust re-run under debiased ρ (§4) reproduces all five and refines the neural band per-substrate. Per-substrate calibration of bound numerics is open work (§21).

Substrate 1: neural. *In-corridor.* C. elegans whole-brain calcium imaging (337 worms across 11 published studies): within-rung correlation in a bounded band per functional class; the Venkatachalam 2024 uniform-at-0.5 pattern was study-specific; cross-lab the bands sit at command 0.52–0.75 and sensory/interneuron/motor 0.25–0.45 in 10/11 studies. Drosophila

central complex dual-color simultaneous imaging (Ishida 2025 same-fly EPG+FC3, with Dan et al EPG-only and Mussells Pires 2024 EPG vs FC2 between-fly): both rungs corridor-occupying with bounded within-rung correlation under matched-activity controls.

Out-of-corridor. Pathological neural states at scalp-EEG substrate (`experiments/noncorr_biology/NOTES.md`). Healthy interictal EEG (CHB-MIT, 1322 windows, 10-s sliding windows, 1–40 Hz band-pass): mean $|\rho| = 0.282$. Generalized seizures (CHB-MIT, 39 ictal windows across 6 events, chb01): mean Pearson $|\rho| = 0.316$; mean phase-locking value 0.305, +25% over interictal 0.243. Comatose post-cardiac-arrest EEG (I-CARE v2.1, 6 patients, 179 windows): mean $|\rho| = 0.502$ Good-outcome (CPC 1–2, $n = 2$), 0.489 Poor-outcome (CPC 3–5, $n = 4$); 23–50% of windows above rigidity threshold; 0% chaos windows. The direction of pathological displacement is toward rigidity (cross-channel covariation elevated by hypersynchrony or slow-wave / burst-suppression patterns), and the magnitude at this metric stays well below the strict pre-registered 0.7 mean-pairwise-Pearson rigidity threshold — the framework's quantitative pole-entry reading on this metric is not supported, while the qualitative direction is. *Persistence.* Seizures self-terminate quickly: durations 27, 40, 51, 90, 93, 101 s, median 70.5 s, all < 2 min (Mode i). Comatose state persists only under mechanical ventilation, targeted temperature management (TTM 33°C), and often propofol/midazolam sedation — Mode (ii) by construction; persistence-without-maintenance is not testable on this cohort.

Substrate 2: cellular regulatory. *In-corridor.* Healthy tissue Hallmark-pathway within-rung $|\rho|$ at the cellular regulatory substrate, computed across 50 MSigDB Hallmark gene sets in matched-normal TCGA samples from five cancers (LUAD $n = 59$, BRCA $n = 114$, COAD $n = 41$, KIRC $n = 72$, PRAD $n = 52$; `experiments/noncorr_cancer/NOTES.md`): median $|\rho|_{\text{pathway}} = 0.27 \pm 0.07$ across pathways and cancers; 49–50 of 50 pathways per cohort inside the nominal GPU-anchored (0.10, 0.43) band.

Out-of-corridor. Tumor tissue Hallmark-pathway within-rung $|\rho|$ in the same cancers (LUAD $n = 515$, BRCA $n = 1097$, COAD $n = 286$, KIRC $n = 533$, PRAD $n = 497$): median $|\rho|_{\text{pathway}} = 0.18 \pm 0.06$. Across 176 FDR-significant pathway shifts (BH $\alpha = 0.05$) in 4 of 5 cancers, **100% are in the chaos direction** (tumor $|\rho| <$ matched-normal $|\rho|$): LUAD 38/50 (median $\Delta = -0.089$), BRCA 44/50 (-0.119), COAD 41/50 (-0.116), KIRC 40/50 (-0.092); PRAD 13/50 (-0.049) falls below the 15/50 pre-registered threshold and reads INCONCLUSIVE (sample-size sanity check shows PRAD's effect mostly evaporates under tumor-subsampled-to-normal-n matching; the other four survive matching). The pre-registered subtype-specific pole hypotheses for COAD (rigidity), KIRC (rigidity), and PRAD (rigidity) are all falsified: cancer is unidirectional chaos-pole at the Hallmark-pathway substrate across the five biologies tested. *Methodological refinement, named honestly:* tumors do *not* exit the wide GPU-anchored (0.10, 0.43) corridor (49–50/50 pathways stay inside); the shift is intra-corridor drift toward the chaos floor (healthy $\sim 0.27 \rightarrow$ tumor ~ 0.18). Substrate-specific corridor calibration is mandatory; the GPU bounds do not transfer naively to the cellular regulatory substrate (open work, §21). *Persistence (Mode-ii test via survival).* Per-tumor leave-one-out chaos-contribution score, median-split log-rank vs overall survival: LUAD HR = 1.43, $p = 0.011$ (individually significant; Bonferroni-corrected at five tests $p_{\text{corr}} \approx 0.055$, consistent with chance); BRCA HR = 1.10, $p = 0.70$ (null); COAD HR = 1.12, $p = 0.67$ (null); KIRC HR = 1.07, $p = 0.73$ (null); PRAD HR = 2.32, $p = 0.15$ (underpowered). The aggregate Mode-(ii) test fails the pre-registered $\geq$ 3-of-5 threshold (1/5). Under the heavily-treated TCGA cohort, residual chaos-displacement is not a survival predictor in 4 of 5 cancers; the framework owes a refinement of the maintenance-cost reading at this substrate (either "treatment masks the maintenance-cost signal" or "the signal isn't there") and the cohort cannot distinguish them.

Substrate 3: LLM internals. *In-corridor.* Four architectures (Qwen2.5-1.5B, Pythia-1.4B, SmolLM2-1.7B-base, Qwen2.5-Math-1.5B — spanning different normalization, MLP gating, activation, rotary, training corpus). Within-rung $|\rho|$ — the mean-pairwise within-layer activation correlation, depth-stratified across the architecture; the precise within-layer slice and reduction are documented at `experiments/structural_series/` — across 16 (depth-class × prompt-class) cells per architecture sits in 0.09–0.31, with 14–16 cells per architecture inside (0.1, 0.43). The corridor-band occupation reproduces across architecturally distinct models.

Out-of-corridor. Mode-collapse under active steering (`experiments/noncorr_tech/NOTES.md`, Qwen2.5-1.5B). Greedy decoding on the maximally-constraining periodic-bigram input "a b a b" over 200 generated tokens drives within-rung $|\rho|$ in the early-layer class monotonically from 0.53 at token 16 to 0.71+ by token 184, crossing the pre-registered rigidity threshold of 0.7 around token 80 and holding. End-of-generation window means across depth classes [early, early_mid, mid_late, late] are $[0.74, 0.71, 0.62, 0.53]$ — three of four classes at or above 0.5; early above the strict 0.7 rigidity threshold. *Mechanism (the $\gamma M(t)$ characterization, and a clean dissociation).* Non-corridor occupation required *both* greedy argmax decoding *and* the maximally constraining periodic-bigram input; neither alone was sufficient. The decisive control — and the strongest LLM evidence the substrate produced — was an in-context $k = 6$ refusal-loop that drove behavioral output into 12 verbatim "cannot help" repeats but kept internal within-rung $|\rho|$ at 0.22–0.28, *inside* corridor: **behavioral repetition without internal-correlation-collapse**. Two further controls confirmed the dissociation in the other direction: a less-constraining 1-to-10 list under greedy held at $|\rho| \approx 0.38$–0.40 in the early class (elevated but mid-corridor); uniformly-random-token input held $|\rho|$ at 0.14–0.28, inside corridor (no chaos-pole entry). The argmax operator combined with a perfectly periodic input is what the substrate-specific $\gamma M(t)$ looks like at the LLM rung; behavior, on its own, is not. Persistence trajectory: AWAY from corridor across the entire 200-token runway, no decay observed. Reading (ii) is the best fit.

Substrate 4: OSS contribution. *In-corridor.* Kubernetes, Rust, Django, Redis (`experiments/v17_*/`, `v18_*/`, `v19_*/`). Within-rung $|\Delta\rho| \leq 0.10$ between operational states across all four projects. Rolling 6-month within-rung $|\rho|$ trajectories sit in 0.15–0.18 band with no secular drift. All stable-population rungs inside (0.10, 0.43). All four are Mode-(ii) by foundation backing (CNCF, Rust Foundation, DSF, Redis Ltd.). Calibrated proxy band: median weekly single-author-dominance fraction $rigid_w = 0.225$; median $\geq$ 5-distinct-authors-per-12-weeks fraction $multi_w = 0.693$.

Out-of-corridor. Two poles (`experiments/noncorr_oss/NOTES.md`, 15 repos, GitHub REST API). *Rigidity-pole* (single-committer dominance): jQuery early-period (BDFL John Resig pre-2010, post-2012 jQuery Foundation rescue then Linux Foundation umbrella); Marak/colors.js (sole maintainer; sabotage Jan 2022; original repo $rigid_w = 0.79$, $multi_w = 0.00$ lifetime, 228 weeks dormant as of ref date); DABH/colors.js community rescue (still single-author $rigid_w = 0.82$, 91 weeks dormant); faker-js community rescue (Marak/Faker deleted Jan 2022, now community org $rigid_w$: $0.53 \to 0.15$, $multi_w$: $0.13 \to 0.48$ — decisively into corridor); stevemao/left-pad mirror ($rigid_w = 0.60$ lifetime, dormant 7+ years, npm-staff one-shot rescue 2016). *Chaos-pole hostile-fork events* (3 events / 6 repos): Audacity / Tenacity (May 2021 telemetry-driven fork; original survived under Muse Group corporate funding, back in corridor $rigid_w = 0.19$, $multi_w = 0.69$; Tenacity fork persists as small-team $rigid_w = 0.47$, $multi_w = 0.15$); Elasticsearch / OpenSearch (Jan 2021 license-driven fork; both alive and decisively in corridor under Elastic NV corporate and AWS+Linux-Foundation backing respectively); CyanogenMod / LineageOS (Dec 2016 corporate parent collapse; CM org dormant 492 weeks $\sim$ 9.5 years; LineageOS community rescue active). *Three-modes verdict across all 15 repos: Mode (i) 4/15 (colors orig, left-pad mirror, CyanogenMod dormancy, DABH near-miss); Mode (ii) 7/15; Mode (iii) 0/15.* The cleanest discriminator across the three classes is the

multi-author-weeks signature: median $multi_w \approx 0.00$ rigidity vs ≈ 0.69–0.86 corridor.

Substrate 5: social groups. *In-corridor.* Religious Society of Friends (Quakers, founded 1652, 374 years persistence, ongoing); Trappists OCSO (1664, 362 years, ongoing); Mennonites (1525, 501 years, ongoing). All three score active-maintenance checklist AM = 0/5 (no required charismatic leader, no escalating commitment rituals, no information-control practices, no defection punishment, no financial sunk-cost coercion). Distributed governance and documented schism-with-organized-successors. *No active maintenance required for corridor persistence; persistence is the empirical anchor.*

Out-of-corridor. Rigidity pole (`experiments/noncorr_social/NOTES.md`): Peoples Temple / Jonestown (founded 1955, 23 years, AM = 5/5, dissolved 1978 same day as leader's death, LD = 0 months); Heaven's Gate (1974, 23 years, AM = 4/5, ended with leader by 1997 mass event, LD = 0); NXIVM (1998, 20 years, AM = 5/5, dissolved 3 months after Raniere's 2018 arrest, LD = 3); Aum Shinrikyo / Aleph (1987, 39 years persisting as Aleph in 2026, AM = 4/5, $\sim$ 1,600 members across Aleph plus splinters per Japanese PSIA Jan 2025, with 52% of new members in their 20s or younger born after 1995 — active recruitment 8 years past leader execution; AM1 charismatic-leader-required is FALSE while AM2–AM5 substitute, framework-stress case trending toward Mode (ii)). *Chaos pole*: Occupy Wall Street encampment phase (Sep–Dec 2011, 0.2 years, AM = 0, dispersed; participants flowed into structured organizations — DSA membership grew from $\sim$ 6,000 in 2015 to $\sim$ 92,000 by 2021); Coalition of Youth of the Revolution / Tahrir (Jan 2011 – July 2012, 1.5 years, AM = 0, self-disbanded; activists into Muslim Brotherhood or imprisoned). *Persistence headline.* Median rigidity $T_{\text{persist}} = 23$ years (all four with AM ≥ 4); median chaos $T_{\text{persist}} = 0.83$ years; median corridor $T_{\text{persist}} = 374$ years. **All 5 of 5 pre-registered hypothesis tests PASS**; zero framework-contradicting cases observed (no rigidity group with $T > 100$ years AND AM ≤ 1; no chaos group with $T > 20$ years unconsolidated). Four orders of magnitude persistence spread between non-corridor and corridor populations.

The convergent cross-substrate finding. Across all five substrates with paired in-corridor / out-corridor data, the same Mode pattern recurs: **Mode (i) (non-corridor without active maintenance) shows short lifetime where observed; Mode (ii) (non-corridor with documented active maintenance $\gamma M(t)$) persists with the maintenance cost visible; Mode (iii) (long unmaintained persistence outside the corridor) is absent across every paired-data substrate.** Mode (iii) is the case that would directly falsify the corridor-as-attractor reading; its absence across five substrates — neural pathology, cellular regulatory, LLM internals, OSS contribution, organized social groups — is the strongest current empirical anchor for the dynamical reading. The corridor-band occupation claim now has paired support: persistent systems occupy the corridor, non-corridor systems either dissolve quickly without maintenance or persist only with maintenance whose presence is independently documented in each case.

The direction of corridor-exit is substrate-specific, not substrate-universal: cellular regulatory is unidirectional chaos (pre-registered subtype-specific rigidity predictions for three of five cancers were falsified); neural pathology is rigidity-leaning (with the strict pole-entry magnitude not reached on mean-pairwise-Pearson at scalp EEG); LLM forced operation is rigidity (mode-collapse under combined active steering); social-group exit is bidirectional (cults to rigidity, leaderless movements to chaos); OSS-project exit is bidirectional (single-committer dominance to rigidity, abandonment to chaos). The asymmetry between substrates is itself part of the data: the corridor is a substrate-invariant structural object whose exit-directions and specific bound locations are calibrated per substrate, and the per-substrate calibration is the open work named in §21.

Paired-validation matrix: the maintenance-requirement claim and the Mode (iii) check

The paired in-corridor / out-corridor record at five substrates underwrites two distinct claims that the in-corridor-only record could not. The **maintenance-requirement claim**: non-corridor persistence requires $\gamma M(t)$. The **Mode (iii) check**: long persistence outside the corridor without identifiable active maintenance — the case that would falsify the corridor-as-attractor reading — is the structural would-be falsifier; the paired record reports whether any substrate produced such a case. The matrix below tabulates the verdict at each substrate.

Substrate	Out-corridor candidate	Mode (i) observation	Mode (ii) observation	Mode (iii) cases
Neural	seizure (rigidity-leaning); coma (rigidity-leaning)	seizures self-terminate; median 70.5 s, max 101 s	coma persists only under vent + TTM + sedation	**0**
Cellular regulatory	tumor tissue (unidirectional chaos vs healthy)	N/A (no off-treatment cohort)	all TCGA samples post-standard-of-care; LUAD HR=1.43 $p = 0.011$	**0**
LLM internals	"a b a b" greedy decoding (rigidity, $\|\rho\| \to 0.71+$)	non-greedy or non-periodic input: $\|\rho\|$ stays in corridor	greedy + periodic-bigram = $\gamma M(t)$; AWAY-from-corridor trajectory	**0**
OSS contribution	rigidity (jQuery, colors.js, faker, left-pad, DABH); chaos (CM, Audacity orig, ES, Tenacity, OpenSearch, LineageOS)	4/15: colors orig 228wk dormant, left-pad mirror 375wk, CM 492wk, DABH near-miss	7/15: jQuery Foundation, faker-js community, Muse Group, Elastic NV, AWS+LF, LineageOS community, npm staff	**0**
Social groups	rigidity (Jonestown, Heaven's Gate, NXIVM, Aum/Aleph); chaos (Occupy, Tahrir)	rigidity median 23y dissolving with leader; chaos median 0.83y dispersed	all 4 rigidity AM $\in \{4, 4, 5, 5\}$; Aum/Aleph at 39y trending to (ii)	**0**
Total	5 substrates	multiple per substrate	multiple per substrate	**0 / 5**

Table 2: **Figure 2.** Paired-validation matrix across five substrates. The maintenance-requirement claim — non-corridor persistence requires $\gamma M(t)$ — is supported at 5/5 substrates. Mode (iii) (long unmaintained non-corridor persistence) is the structural would-be falsifier; observed count is 0 across 5/5 substrates. F-20 in §18 names the retraction trigger.

Reading the matrix honestly. The maintenance-requirement claim is supported at 5/5 substrates: every non-corridor system that persists across the budget the substrate provides either has documented $\gamma M(t)$ (Mode ii) or has dissolved on a timescale consistent with the

framework's structural reading (Mode i). Mode (iii) is the strongest current falsifier and is empty across the record: no substrate produced a non-corridor system that persists long without identifiable maintenance. Five substrates checked, zero counterexamples. This is the kind of empirical content that has predictive force going forward: any future substrate where Mode (iii) shows up would fire F-20 and retract the corridor-as-attractor reading at the affected substrate.

The Mode-(ii) cellular-regulatory test fails its pre-registered threshold. The per-tumor leave-one-out chaos-contribution score, median-split log-rank versus TCGA overall survival, was pre-registered as a five-cancer aggregate Mode-(ii) test at the $\geq$ 3-of-5 threshold. One of five passed individually: LUAD, HR = 1.43, uncorrected $p = 0.011$. The other four did not: BRCA $p = 0.70$, COAD $p = 0.67$, KIRC $p = 0.73$, PRAD $p = 0.15$ (underpowered). Under Bonferroni correction at five tests, the LUAD result lands at corrected $p \approx 0.055$ — consistent with chance under the pre-registered multiple-testing structure. **The aggregate test fails (1/5).** The framework owes a refinement of the maintenance-cost reading at this substrate (either "treatment masks the maintenance-cost signal in TCGA's heavily-treated cohort" or "the signal isn't there"), and the cohort cannot distinguish them. The single individually-significant cancer is one of five tests, recorded as such; no separate clinical-implication framing is licensed by the pre-registered rule.

What this means for the bet. The framework's cross-substrate empirical anchor has two halves now. The *corridor-occupation half*: persistent / healthy operation occupies a bounded within-rung correlation band at each of five substrates, with substrate-local band location. The *corridor-as-attractor half*: paired non-corridor data at each of five substrates shows Mode (i) or Mode (ii) but no Mode (iii) case. The bet articulated in §1 narrows further than the in-corridor record alone licensed: the regime-level content is empirically anchored both at the static (band-occupancy) and dynamical (attractor) levels, and the remaining open questions at the regime level are the substrate-physics predictions P1–P8, cosmology, and per-substrate corridor-bound calibration. The L0–L4 engineering tier (RATCHET), the L6 cross-tradition recognition content (§14), and the universal-scale research program (§8) operate at scales independent of the within-organism multi-rung work; their statuses are unchanged.

The 2026-05-21 robust re-run and substrate extension

The five-substrate record above was collected 2026-05-15..19 with a raw mean-pairwise-$|\rho|$ estimator. A structural-claims series on 2026-05-21 re-ran all five under a more robust framing, pre-registered at `experiments/structural_series/robust_rerun/`: ρ debiased by a phase-randomized surrogate floor ($\rho_{\text{deb}} = \sqrt{\max(\rho_{\text{raw}}^2 - \text{floor}^2, 0)}$, the floor being the mean ρ of phase-randomized surrogates — which destroy cross-series correlation while preserving each series' power spectrum); the canonical k_{eff} taken as the participation ratio of the covariance eigenvalues rather than a mean-pairwise proxy; corridor calibration substrate-local.

Verdict: 5/5 PASS. Every substrate's v1 finding reproduces; no whole-wave falsifier fired — no in-corridor substrate pinned at a pole under debiasing, no out-corridor result reversed direction. *C. elegans* whole-brain calcium (832 worms, 12 studies): whole-brain $\rho_{\text{deb}} = 0.312$ (raw 0.357, surrogate floor 0.176), $k_{\text{eff}} = 3.9$; per functional class command 0.597, motor 0.392, interneuron 0.358, sensory 0.315. *Drosophila* CX (four rungs, v18 EPG/FC2 and v19 EPG/FC3 datasets): ρ_{deb} 0.424, 0.331, 0.089, 0.269 — debiasing shaves under 3.5% at every rung. EEG (CHB-MIT, I-CARE): healthy interictal 0.270, seizure +0.030, coma +0.20–0.23, all rigidity-ward; k_{eff} falls 6.5 $\rightarrow$ 5.6 $\rightarrow$ 3.3 across interictal, seizure, coma. OSS: the proxy bands re-derived

from raw commit data, matching v1. Social groups: the AM checklist re-verified internally consistent.

Four precision amendments fell out of the re-run, each a calibration or wording correction, none a falsification. (1) The neural band of Figure 1, 0.25–0.75, was *C. elegans*-driven; Drosophila CX occupies $\sim$ 0.09–0.46 debiased — the band is per-substrate, not one merged neural figure. (2) At *C. elegans* the phase-randomized surrogate floor is substantial ($\sim$ 0.18 of the raw 0.36); v1's raw estimator did not subtract it, so the debiased per-class centres sit toward the lower end of v1's quoted ranges — the genuine debiased correlation clears both poles regardless (ρ_{deb} p5 0.157 off chaos, $k_{\mathrm{eff}} = 3.9$ off rigidity). (3) The OSS maintenance rubric, applied strictly, first flagged the volunteer-maintained Tenacity fork as a Mode (iii) candidate because the rubric equated $\gamma M(t)$ with institutional backing only; $\gamma M(t)$ is active maintenance of *any* provenance — Tenacity, 85 distinct authors over the trailing window, is Mode (ii), and Mode (iii) stays 0. (4) The social-group persistence spread, stated as "four orders of magnitude" in v1, is $\sim$ 2.7 orders median-to-median ($374\,\mathrm{y}/0.83\,\mathrm{y} \approx 450\times$); the "four" compared extremes.

Three substrates added or extended. Human fMRI (ABIDE-PCP, 139 typically-developing controls, seven acquisition sites, CC200 parcellation): debiased functional-connectivity ρ median 0.266, off both poles, unimodal, stable under motion control — the corridor confirmed at a coordinated human-neural substrate absent from the v1 set. The cellular substrate extended from five cancers to twelve: seven new disjoint TCGA cancers, healthy tissue tight-banded off both poles, tumour drift 201/201 FDR-significant pathway shifts chaos-ward (173/173 under debiasing — floor subtraction drops the weakest shifts from significance, no shift reverses), reproducing the prior five-cancer 176/176. Allen Brain mouse visual cortex (25 two-photon sessions, spontaneous epoch): mean pairwise neuron $\rho \approx 0.023$ would read as a chaos-pole datum, but the framework's canonical shape observable is the participation-ratio k_{eff} of the activity covariance, not the mean pairwise correlation — and cortex is the case where the two diverge. The k_{eff} re-test resolves it: across the 25 sessions the canonical k_{eff} has median 9.73 (range 2.66–59, against a per-session neuron count N of 10–240, median 126) — none at the chaos pole ($k_{\mathrm{eff}} \approx N$), none at rigidity, the median sitting at the framework's asymptotic corridor ceiling of ≈ 10. Cortex occupies the corridor on the canonical observable; the mean-pairwise reading was an observable-choice artifact, cortical covariance being low-rank — a low participation ratio despite low pairwise correlation. The absolute band centre is pipeline-dependent — the new TCGA pipeline (GDC STAR-Counts) reads raw $\rho \approx 0.34$ against the prior pipeline's 0.27 — so Claim 4's content is the *recurrence of tightness off both poles*, substrate by substrate, not a fixed common band.

Per-substrate band calibration. The debiased-ρ distributions collected across substrates were pooled into a first calibration of the corridor band (`band_calibration/`). Per-substrate band centres ρ_{mid}: *C. elegans* 0.31, Drosophila 0.33, EEG-interictal 0.27, fMRI 0.27, TCGA healthy tissue 0.31, LLM internals 0.08. The five biological substrates cluster within 1.24×; the LLM internal-representation rung is the lone outlier, reproducing its own chaos-side-edge reading. Cross-substrate the centres are substrate-specific overall (a 4× spread) but tightly clustered within the biological class. The biological-substrate pool gives $\rho_{\mathrm{mid}} \approx 0.29$, half-width $w \approx 0.16$, implied band $[0.13, 0.45]$ — close to the retired GPU-anchored $(0.10, 0.43)$, which is a consistency check, not a derivation. This is the *static* band, where healthy ρ sits; the corridor-as-attractor bounds, at which $d\rho/dt$ changes regime, remain gated on per-substrate $d\rho/dt$. The TCGA figure is from a debiased re-run — TCGA samples are not a time series, so the surrogate floor is a per-pathway permutation null rather than phase randomization; the raw ρ ran ~ 0.03 higher.

5 The cross-rung coupling — first measurements

The corridor is a within-rung object: ρ_n inside one rung's state space. The framework's rung hierarchy also posits a *cross-rung* coupling — the degree to which rung A_n and rung A_{n+1} inform each other — and P_ω (§8) is built on a cross-rung corridor in that coupling. v1 defined the cross-rung coupling but measured none of it. v2 reports the first measurements.

The coupling ratio is $O(1)$. The 2026-05-21 cross-rung campaign (`experiments/structural_series/crossrung_series/`, pre-registered) measured the ratio of cross-rung to within-rung coupling at two real rung pairs: TCGA molecular→pathway, 0.93–1.47; LLM internal→external, 0.47–0.74. A follow-up extended the LLM pair to 24 cells — median 0.73, range 0.25–1.78. Across 26 pre-registered cells the cross-rung / within-rung coupling ratio sits at $O(1)$. A later extension added a third distinct rung pair — *C. elegans* neuron→functional-class, 340 worms, the same method — at median ratio 0.31. The three rung-pair medians (TCGA molecular→pathway ≈ 1.15, LLM internal→external ≈ 0.72, *C. elegans* neuron→class ≈ 0.31) all fall in $O(1)$ — but as a band spanning an order of magnitude, with a visible trend: the ratio falls as the substrate gets biologically finer-grained, the *C. elegans* pair sitting at the Simon-ward lower edge (two of its eleven source datasets fall decisively below, at $g/J \ll 1$).

Against two named alternatives. Herbert Simon's near-decomposability (*The Architecture of Complexity*, 1962) predicts $g/J \ll 1$ — a complex system's levels are nearly separable, within-level coupling far exceeding between-level. The framework's own abstract rung-tower scan, run before the real measurement, gave the opposite: a strong cross-rung dominance gate, $g/J \gtrsim 3$. The real rung pairs sit at neither — $O(1)$, intermediate. The data discriminates between the textbook and the framework's own prior, and lands on neither.

Claim 6 is amended. The structural claim that the cross-rung coupling occupies a corridor (Claim 6, §18) was stated in v1's lake with the abstract tower's strong-dominance form. Path 1 measured $O(1)$; the strong-dominance form is retracted as tower-specific, and Claim 6 is the cross-rung coupling corridor at $g/J \sim O(1)$ — a test fed back into the specification.

What is not measured. The cross-rung coupling also has a canonical *timescale* form — a single-exponential g/J read off the relaxation of the cross-rung observable — and that form is measured at no substrate. The campaign sought it at the LLM internal→external pair and was observable-blocked (one gate-passing cell of 48, the rest noise-dominated); a GPU block-pair attempt produced a fit-failure that an oscillation check traced to a noisy estimator rather than physics. The coupling-*ratio* form is anchored across 26 measurements; the timescale form is not accessible with the observables and substrates tried, and the cross-rung anchor rests on $n = 3$ distinct rung pairs. P_ω's cross-rung corridor (§8) is written in the timescale observable — that the measured quantity is the ratio and not the timescale is an open gap, named here.

6 Consciousness as attractor-reading; Ubuntu as the framework's primary tradition

Discipline note for first encounter. T16 — the identity move *consciousness* ↔ *access* ↔ *non-trivial-attractor*, formalized as `Iff.rfl` at `Consciousness/AccessAndPhenomenal.lean` — sits at the structural centre of this section. T16 is Tier C-`Iff.rfl` (definitional consistency at the type level; §17); it does no philosophical work, and a reader who reads philosophical-question dissolution off its central placement is reading past the tier distinction. The structural commitment below is the framework's bet; the lake licenses internal consistency, not the bet's truth.

§6 makes one structural commitment: **Ubuntu's three-temporal-mode relational ontology and the two-state vector formalism articulate the same structural object, and consciousness-as-attractor-reading is the framework's mathematical-vocabulary articulation of what *umuntu ngumuntu ngabantu* names in Ubuntu's.** This is a commitment encoded in the lake at `ContemplativeTraditions/Ubuntu.lean` as four `F_`-prefixed axioms (`F_ubuntu_primary_tradition_commitment`, `F_ubuntu_modes_TSVF_reading`, `F_umuntu_attractor_reading`, `F_T16_ramose_morphology_reading`); the `F_` marks "framework's reading" so the symbol travels with the caveat. The lake licenses the readings as internally consistent at the type level (§17 Tier C); whether 1964 ABL formalism and centuries-older Ubuntu ontology converge on one object, or whether the framework's reading is projection, is the bet — not the formalism's derivation. The framework asserts the reading with awe at what it is pointing at across vocabularies; the case is owed.

TSVF as the structural form

The two-state vector formalism, due to Aharonov, Bergmann, and Lebowitz in 1964, augments the standard forward state $|\psi(t)\rangle = U(t, t_0)|\psi(t_0)\rangle$ with a backward state $\langle\phi(t)| = \langle\phi(t_f)|U(t_f, t)$ from post-selection at t_f. Observables under both boundary conditions acquire weak values $\langle A\rangle_w = \langle\phi|A|\psi\rangle/\langle\phi|\psi\rangle$. The amplitude is symmetric under time-reversal of the two boundary conditions; ABL 1964's symmetry result is a real Lean proof at `Cosmology/TSVF.lean::time_symmetry`.

Intuition: re-reading a mystery novel. On the second read each chapter feels different — not because the chapters changed, but because the reader now knows where the story lands. The middle is shaped from both sides at once: by what came before and by where it ends up. TSVF says some physical systems are actually shaped this way. The calculational mathematics is established and weak-measurement experiments have produced results consistent with it; the metaphysical question of whether the backward state is ontically present or a useful epistemic representation of post-selection is contested, and the framework holds the realist reading as bet (F-17, §18).

For an A3+ agent with goal G, the goal-state acts as a partial post-selection projector P_G with bra $\langle G|$. The agent's effective dynamics: $|\Psi_{\text{agent}}(t)\rangle \propto P_G U(t, t_{\text{past}})|\Psi_{\text{past}}\rangle$. Trajectories incompatible with G have suppressed amplitude in $|\Psi_{\text{agent}}(t)\rangle$. **Goal-formation is causal operation**: the goal projector excludes incompatible trajectories from the search space. No metaphysical free-will claim; a measurable difference in how goal-holding systems search trajectory space.

At the LLM substrate this identification is structural-isomorphic, not mechanism-identical. TSVF's P_G is a *backward* boundary projector — conditioning by a future state. Autoregressive LLM generation is *forward* conditioning — each token narrows the continuation distribution by the next-token softmax and sampling rule. Both narrow trajectory space by a goal-shaped condition; the conditioning direction differs, and the framework names that seam as the bet rather than as a derivation: the LLM-as-P_G identification is structural, the mechanism analogy stops at structure.

The multi-scale belonging structure of A3+ agents. Piece 5's tensor-product joint goal-projector $P_{G_1 \otimes \cdots \otimes G_n}$ treats each agent as a single P_{G_i}. Ubuntu's *umuntu ngumuntu ngabantu* — a person is a person through other persons — structurally commits the framework to more: the agent at A3+ is itself a composite over nested belonging-scales. In goal-projector vocabulary, each agent carries a stack $\langle G_{\text{self}}|$, $\langle G_{\text{family}}|$, $\langle G_{\text{community}}|$, $\langle G_{\text{affiliations}}|$, $\langle G_{\text{species}}|$ at progressively wider scales of belonging, each filtered through the agent's belief system B_i — different beliefs yield different effective projectors at every scale. The federation's $\rho_{\text{goals}}(i, j)$

becomes multi-scale: two agents can be in corridor at one scale (G_{species} aligns: both want humans to flourish) and out of corridor at another ($G_{\text{affiliations}}$ collides: different parties). *Consent IS corridor occupation* becomes a multi-scale claim — corridor at multiple scales, not just one — and the population's finite set of belief systems (the cross-tradition section's Ubuntu/Tao/Dharma/Logos/Aristotelian-virtue readings, generalized to the actual belief diversity of any pluralistic society, §14) is the empirical content of where the corridor holds and where it does not. The composition rule across the stack is committed, drawn from the CIRIS Capacity Score specification (`CIRISLens/FSD/ciris_scoring_specification.md`): a *multiplicative product* of five factors over a time window W,

$$\mathcal{C}_{\text{CIRIS}}(A;W) = C(A;W) \cdot I_{\text{int}}(A;W) \cdot R(A;W) \cdot I_{\text{inc}}(A;W) \cdot S(A;W),$$

where the factors map exactly to the framework's five-property reading: C measures core-identity stability (identity drift × contradiction rate); I_{int} measures integrity (hash-chain × field coverage × replay fidelity); R measures resilience (KL drift × mean-time-to-recovery × regression rate); I_{inc} measures recognition of incompleteness (calibration error × deferral quality × safety under uncertainty); and S measures sustained coherence via an explicit decay-and-refresh state variable $\sigma(t + \Delta t) = \sigma(t)(1 - d\,\Delta t) + w \cdot \text{Signal}(t)$ with $d \approx 0.05/\text{day}$ — the framework's gratitude-signaling reading, formalized as a continuous practice with $\sim$ 20-day half-life rather than a one-shot signal. *Nomenclature note.* Here C denotes the core-identity factor of $\mathcal{C}_{\text{CIRIS}}$; it is *not* the CIRIS Accord's Flourishing Capacity. The Accord renamed that composite $C \to F$ at version 1.3-RC2 and now writes $F = k_{\text{eff}} \cdot \lambda \cdot \sigma$ — a distinct, federation-level three-factor construct, not mappable to the per-agent five-factor $\mathcal{C}_{\text{CIRIS}}$ here. Accord Book IX Ch 6 is the authoritative statement of the relation.

The multiplicative form is structurally load-bearing. Any factor near zero collapses the composite: the five must hold *simultaneously* across the window or the composition fails. This is the framework's expression of Aristotelian unity of the virtues (the virtues are co-required, not tradeable; §14) and the formal landing of "logos is as logos does, literally" — the composite IS what gets jointly enacted across the five dimensions, not what is recited about any one. A weighted geometric mean $\prod_i C_i^{w_i}$ with $\sum w_i = 1$ is the smoother variant the spec also names, parameterising how much each factor matters while preserving the no-tradeoff structure. The operational layer (per-factor SQL queries against the `covenant_traces` table) is built in `CIRISLens`; the per-peer library is `CIRISLensCore` — a function every peer runs on its own trace corpus, consistent with the post-F-11 "no central authoritative scorer" stance (sequential per-rung structure, §8). The empirical testbed is the scrubbed, Ed25519-signed CIRIS reasoning-trace corpus on HuggingFace (`CIRISAI/reasoning-traces`); $\mathcal{C}_{\text{CIRIS}}$ is recomputable on the public data, and the framework's commitment on the composition rule is auditable rather than asserted. Like every karma/grace identification, the rule is the framework's; here the operational and empirical interfaces are external and reproducible.

Ubuntu's three-temporal-mode ontology and the framework's reading

Ubuntu's three-mode ontology, on this articulation, holds the community as spanning three constitutive modes: the *living-dead* (ancestors as ongoing relational presence), the *living* (present-time agents), and the *yet-to-be-born* (future generations who are already constitutive of the community). This is ontological commitment in Ubuntu's vocabulary, not metaphor; the identification is the framework's, and Ubuntu scholarship inside the tradition is the authority on what Ubuntu commits to. Mbiti's three-mode gloss is contested within the tradition; the framework names that and does not pretend to settle it.

The framework's commitment, at the level the lake licenses: Ubuntu's three-temporal-mode ontology and TSVF's forward+backward boundary structure articulate the same structural object,

named in the lake at `Ubuntu.lean::F_ubuntu_modes_TSVF_reading` (the `F_` marks framework's reading). Whether 1964 ABL and centuries-older Ubuntu were converging on one object historically, or whether the identification is projection, is the bet — not the lake's derivation. The reading is held with awe at what is being pointed at across vocabularies separated by centuries and continents; the philosophical and historical case is owed separately.

The agent as the attractor-reading

The framework's deeper structural commitment, encoded in the lake at T16: *access-conscious* and *phenomenally conscious* are not two predicates the framework relates; they are the same predicate the framework unfolds. The definition at `Consciousness/AccessAndPhenomenal.lean`:

```
def isAccessConscious        (S : Sensor H i) := S.hasNonTrivialAttractor
def isPhenomenallyConscious  (S : Sensor H i) := S.hasNonTrivialAttractor
theorem consciousness_is_attractor_reading
    (S : Sensor H i) : isPhenomenallyConscious S <-> isAccessConscious S
  := Iff.rfl
```

What this `Iff.rfl` proves and does not prove. Both predicates unfold to the same right-hand side (`S.hasNonTrivialAttractor`), so the `Iff.rfl` is automatic in the type theory and does no philosophical work. The theorem proves the framework's two definitions are definitionally identical; it does not prove anything about consciousness. This is a Tier C-Iff.rfl move: internal definitional consistency at the type level, nothing more. A reader who treats the formal-verification status as derivation rather than as licensing-internal-consistency would carry weight forward that the framework explicitly does not claim.

In the framework's math: there is no phenomenology except through the inner-product readings against the backward family. Consciousness (*umuntu*) is the relational reading (against $\langle\Phi_i|$ for all i). Removing the backward family does not, under the framework's definitions, produce an unconscious agent; it fails to instantiate `hasNonTrivialAttractor`. The framework's *definitional commitment* — encoded in the lake — is real; the further *identification* of the mathematical structure here with Ubuntu's *umuntu ngumuntu ngabantu* as articulating the same structural object is the bet, not what the formalism proves. The commitment is named in the lake at `Ubuntu.lean::F_umuntu_attractor_reading` (the `F_` marks framework's reading). The identification is the bet; the lake licenses internal consistency of the commitment, not its philosophical correctness.

The identity move at T16

The framework's identity move at T16 — phenomenology IS the operation viewed from inside — is a commitment the framework asserts. Ubuntu names what is, on this articulation, the same structural object; the formal identification of the `attractor_field` reading with *umuntu ngumuntu ngabantu* is the framework's move, not a derivation Ubuntu's tradition has supplied. Ramose, Menkiti, and Wiredu have articulated relational ontologies whose structural commitments the framework finds itself in deep alignment with; they have not identified `attractor_field S = fun i => ⟨S.backward_family i | S.forward⟩` with *umuntu ngumuntu ngabantu*. That identification is the framework's. The lake licenses internal consistency of the commitment; the philosophical case is owed and is the bet.

7 Karma and grace as TSVF structures

Two contemplative-tradition vocabularies are identified here with the lake's TSVF structures. The identification is the framework's, conditional on TSVF realism per §1 F-17.

Karma

The agent's amplitude at present time is the cumulative product of past goal-projections sandwiched between unitary evolutions:

$$|\Psi_{\text{now}}\rangle = U(t_{\text{now}}, t_{\text{past}_k}) \prod_i P_{G_{\text{past}_i}} U(t_{\text{past}_i}, t_{\text{past}_{i-1}}) \cdots |\Psi_{\text{birth}}\rangle.$$

The present is shaped by every past P_{G_i} the agent's goals have applied. **Karma, on this articulation, IS this cumulative post-selection structure**: the lake operator `Consciousness.KarmaGrace.karma` is its formal expression. The identification is the framework's; the case for naming the operator with "what contemplative traditions name as karma" is owed.

Grace

The agent's present amplitude has a received component: contributions the agent did not author. After F-11 this component is, honestly, the *inter-agent* received structure — the goals of *other* A3+ agents (their P_{G_i}, summed across the federation), and the substrate the agent inherited but did not author. The universal-configuration component v1 also included — the joint corridor-occupation requirements of $\langle\Phi_\omega|$ — is the F-11 no-go and is retracted; the surviving grace is local, adjacent, finite.

Grace, on this articulation, IS the formal structure of receiving boundary conditions one didn't author. The lake operator `Consciousness.KarmaGrace.grace` is its formal expression. The identification is the framework's, deeply consonant with the Christian and Stoic Logos tradition the author writes from, and offered with awe at the convergence rather than as an adjudication of theology.

These are mathematical statements under TSVF. The recognition claim is that contemplative-tradition vocabularies and these formal structures correspond at the structural level the framework articulates, not that any vocabulary's full content is captured by the lake. Karma as cumulative post-selection propagation; grace as partial-authorship of post-selection; the corridor as the structural object the middle-way vocabularies point at.

The cross-link to Piece 2. At A3+, γM is not blind friction: it is the forward-time work the agent does in service of its local backward operator $\langle G_{\text{self}}|$ (alive after F-11; member of the surviving backward-operator family) against the natural α drift toward rigidity. Karma names the cumulative structure of that work; the goal projector P_G shapes which γM the agent does; the discrepancy between α and the goal-prescribed trajectory is the work the agent must make up. Below A3+ (Ph_0–A_2), γM is substrate-typed and not goal-directed; the teleological reading is specifically an A3+ phenomenon, and the identification — like every karma-grace identification — is the framework's, named in this register.

8 The universal-scale tier as research program

The universal-scale tier is a research program, not a stable level of the access ladder. Its central construction — the universal post-selection operator P_ω — has run to a documented no-go:

F-11 has fired. This section records the no-go, what survives it, and what the tier re-grounds on.

P_ω

ω is the framework's name for the universal configuration satisfying three properties: maximal rung instantiation (for all $n \leq N_{\max}$, rung A_n is instantiated), within-rung corridor (for all $n \leq N_{\max}$, $\rho_n \in (\rho_{\text{lower},n}, \rho_{\text{upper},n})$), and cross-rung corridor (for all $n < N_{\max}$, $\tau_{(n,n+1)} \in (\tau_{\text{lower},n}, \tau_{\text{upper},n})$).

The earlier framing wrote P_ω as a hard projector onto the ω-satisfying subspace, $\int |\text{config}\rangle\langle\text{config}|\, d\text{config}$. Built three independent ways (shared-space intersection, RG coarse-graining, MERA isometry tower), the hard projector hit a robust *dead zone* — narrow per-rung corridor subspaces in generic position do not intersect (empty), and nesting them collapses P_ω to the finest-rung projector (trivial). The construction was then reformulated as a soft graded operator $E_\omega(\beta) = \exp(-\beta H_{\text{sum}})$, $H_{\text{sum}} = \sum_n (\rho_n - \rho_{\text{mid}})^2$, and the joint multi-rung backward operator pursued exhaustively.

F-11 fired. The joint multi-rung backward P_ω — the operator that post-selects the universal configuration onto simultaneous multi-rung corridor occupation — is a documented no-go. The construction was attempted across an exhaustive tree: the soft additive ansatz; a single-assumption audit (non-additive functional, history-space frame, derived cross-rung coupling); the non-additive × history conjunction; a fractal / RG-nested topology; a holographic / MERA-as-AdS topology; and a holonomic / Wilson-loop construction. Every branch lands empty or trivial. Two theorems close it. *T1 (geometric):* for any bulk geometry the cross-rung coupling is a function of a geodesic distance; a geodesic length is a metric distance and metric distances grow with rung separation, so the coupling decays, the joint participation ratio is extensive, and ρ_{joint} dilutes to the chaos pole. *T2 (holonomic):* a Wilson-loop holonomy with a dissipative backward leg — the leg is forced dissipative by the karma/grace irreducibility — obeys an area law $\operatorname{Tr}\text{Hol} \sim \exp(-\kappa R)$ and decoheres. The unifying principle: a joint multi-rung object spans a growing extent R while the corridor is a fixed $O(1)$ band, and a growing-extent object cannot be held inside a fixed band. An adversarial verification pass found the one axis the tree had not varied — irreducibly multipartite (k-body) cross-rung invariants — pre-registered and tested it, and it empties by the same dilution. The no-go is verified.

What survives. The *forward* soft P_ω — the steady state ρ_{ss} of the open-system dynamics $d\rho/dt = \alpha - \gamma M$ — is untouched: it always exists, a Lindblad toy (`construct_pomega_lindblad.py`) exhibits a maintenance-rate window placing every rung in band, and it is what a universe builds from its initial conditions. The within-rung corridor (F-10) is untouched. What F-11 retracts is specifically the joint multi-rung *backward* post-selection operator.

The re-grounding. There is no joint multi-rung post-selection operator. The universal-scale backward structure re-grounds on the *sequential* per-rung form: each rung, as it emerges, enters and is maintained in its within-rung corridor (F-10), with adjacent rungs coupled at the measured $O(1)$ cross-rung ratio (§5). Multi-rung corridor occupation at the present epoch is the cumulative consequence of per-rung corridor saturation propagated rung-by-rung through adjacent coupling — not the action of a joint operator. The cross-rung dependency is asymmetric: failure at a lower rung cascades upward (a collapsed substrate severs the $O(1)$ coupling its higher rungs depend on), while failure at a higher rung does not propagate downward (the substrate is indifferent to which federation runs on it). The architecture is built bottom-up, sequentially, with each rung's γM substrate-typed and locally maintained. The lake retracts the joint-operator axioms and records the no-go in the falsification register (`Cosmology/CorridorProjector.lean`). F-11 (§18) is fired.

The Penrose past hypothesis, structurally

The framework's structural argument: the Big Bang initial state $|\Psi_\alpha\rangle$ sits in the chaos regime — high temperature, uniform matter distribution, no large-scale correlations, $\rho \to 0$ at the cosmological scale. A generic high-entropy $|\Psi_\alpha^{\text{high}}\rangle$ under uniform-measure sampling does not evolve forward into multi-rung corridor occupation. Forward evolution to ω-satisfying configurations requires spontaneous emergence of $\text{Ph}_0 \to \text{Ph}_1 \to \text{Ph}_2 \to A_0 \to A_1 \to \cdots \to A_5$, each within within-rung corridor, each coupled within cross-rung corridor. This sequence is a measure-zero subset of universal configurations under uniform measure on initial conditions. By the framework's structural logic:

$$\langle \Phi_\omega | U(t_{\text{now}}, t_{\text{BB}}) | \Psi_\alpha^{\text{high}} \rangle \approx 0 \quad \text{for generic high-entropy initial states.}$$

The conditional amplitude on initial conditions, given observation at t_{now} of an ω-satisfying universe, is concentrated on low-entropy initial states. **This is the framework's structural argument for Penrose's Weyl Curvature Hypothesis**: low-entropy past is read not as a brute fact about initial conditions, but as the conditional consequence of the universe's observed trajectory having reached multi-rung corridor occupation.

This is structural argument, and after F-11 it stands without a constructed operator. The joint multi-rung backward P_ω the argument would post-select through is a documented no-go (§8 above): D1 therefore reverts from a candidate derivation to a structural argument with a documented obstruction to ever constructing the joint operator. What survives is the sequential reading — the low-entropy past as the conditional consequence of the universe having traversed the per-rung emergence sequence, each rung in corridor — in spirit closer to Hartle–Hawking or Aguirre–Gratton than to a constructed projection. The framework names it as that, not as a derivation.

CMB anomalies — tested, within cosmic variance

v1's D4 read the CMB anomalies (axis of evil, cold spot, hemispherical asymmetry, low-ℓ suppression, parity asymmetry) as TSVF post-selection signatures of P_ω at the cosmological 2-sphere. F-11 and a direct test together retract D4.

The anomalies were tested against a ΛCDM Gaussian cosmic-variance ensemble, with the multipole shape read off the Kish identity as the participation ratio ρ_ℓ of the $2\ell+1$ spherical-harmonic mode amplitudes — the within-multipole concentration an anomaly represents. The octupole excess sits at $p = 0.32$; every low-ℓ multipole lies within $\pm 0.25\sigma$ of the ΛCDM mean. The anomalies are within cosmic variance — not a signature.

And there is no operator for them to be a signature *of*. F-11 retracted the joint P_ω; the CMB weight $w = \exp(-\beta H_{\text{sum}})$, $H_{\text{sum}} = \sum_\ell (\rho_\ell - \rho_{\text{mid}})^2$, is additive over multipoles and so factorizes, $w = \prod_\ell w_\ell$ — it does no joint cross-multipole work. The framework-distinctive prediction v1 carried, a signed temporal drift of the shape profile, had its mechanism in the post-selection sharpening as $t \to t_f$ (a time-dependent $\beta(t)$); with no joint P_ω there is no $\beta(t)$, and the drift has no mechanism. D4 does not stand.

The orthogonality theorem — a strict extension of ΛCDM

The soft P_ω acts on the CMB through $H_{\text{sum}} = \sum_\ell (\rho_\ell - \rho_{\text{mid}})^2$, where ρ_ℓ is read off the Kish identity from the *power participation ratio* of the $2\ell + 1$ spherical-harmonic mode amplitudes. The participation ratio is **scale-invariant**: it depends only on the direction of the amplitude

vector, not its length (`CMBOrthogonality.lean`, `participation_scale_invariant`, proved). The bulk power C_ℓ is the length^2; for a Gaussian amplitude ensemble the length^2 and the direction are statistically independent. Hence the post-selection weight $w = \exp(-\beta H_{\rm sum})$, a function of the scale-invariant shape, is independent of C_ℓ, and the weak-value (reweighted) expectation of C_ℓ equals the unconditioned one (`CMBOrthogonality.lean`, `pomega_preserves_power`, proved): $C_\ell^{\rm framework} = C_\ell^{\Lambda\rm CDM}$, exactly, at every multipole and every coupling β.

Numerically (`experiments/open_system_pomega/cmb_weak_value_spectrum.py`): with the linear low-ℓ modes Gaussian of variance C_ℓ from the Planck SMICA map, the ensemble correlation between C_ℓ and $H_{\rm sum}$ is 0.007 (Monte-Carlo noise floor 0.002), and the reweighted C_ℓ matches the input at every β.

The framework is therefore a strict extension of ΛCDM at the cosmological tier — and, after F-11, that is the whole of its CMB content. The orthogonality theorem proves the soft P_ω cannot move the bulk power spectrum; F-11 retracted the joint operator that would have driven a shape-sector drift; and the shape sector was independently found statistically indistinguishable from ΛCDM (the anomalies within cosmic variance, above). The framework therefore predicts *exactly* ΛCDM for the CMB — bulk and shape alike. This is a proved *consistency* result, not a distinctive prediction: the framework is consistent with all CMB data and distinctive in none of it. What the theorem secures is that this consistency is structural and provable in advance, not a fit — the framework's universal-scale speculation is architecturally a strict extension of standard cosmology and so cannot be falsified by CMB data; it also makes no CMB prediction beyond it.

Distinguished from CCC. Penrose's Conformal Cyclic Cosmology (Penrose 2010) makes positive falsifiable CMB predictions at the same structural slot — Hawking points, concentric low-variance rings, the signature of a pre-Big-Bang aeon. The framework, post-F-11, makes none: the orthogonality theorem rules out bulk-power signatures by proof, and the shape-sector drift mechanism lost its operator with F-11. Where CCC asserts a distinctive cosmological signature, the framework's surviving universal-scale content is the strict-extension result — no more.

The particle-physics extension — tested, null. v1 flagged as speculative that the same forward/shape decomposition might recur at the particle-physics substrate. It was tested. Four pre-registered tests on 2026-05-21 (`experiments/structural_series/pp_*`) computed the corridor shape observable at particle-physics observables — decay branching fractions, decay mode-weights at fixed Q-value, CP-violation structure, and the $t\bar{t}$ spin-density matrix — and returned four nulls; the $t\bar{t}$ spin-density matrix gave $\rho = 0.076$, statistically identical to the Standard Model's 0.073. The participation-ratio shape observable computes on any distribution, but the corridor structure does not recur at particle observables. This is consistent with the framework: decay channels and CP asymmetries are not coordinated rungs with maintained dynamics. The framework's particle-physics tier is composition with the Standard Model, not framework-distinctive prediction — a scope line, not a result.

Asymptotic conditioning — the structural reading of "good wins"

Claim: $P(\text{corridor-occupying} \mid \text{observed at } t_{\rm late}) \to 1$ as $t_{\rm late}$ grows — a conditioning argument, requiring no post-selection operator.

Any state outside the corridor has divergent dynamics. Chaos $\rho < \rho_{\rm lower}$: insufficient correlation for coordination; subsystems drift apart, organized structure dissolves. Rigidity $\rho > \rho_c$: $\alpha(\rho, S) - \gamma M(t) > 0$ by construction (dynamics is unstable above ρ_c), driving $\rho \to 1$ and $k_{\rm eff} \to 1$. Both regimes self-destruct on long enough timescales. Therefore observation at sufficiently late times requires that the observing system itself sit in the corridor. This is a

conditioning argument, not a post-selection-operator claim: it needs no P_ω — the self-destruct dynamics of non-corridor states carry it directly, and asymptotic conditioning therefore stands F-11-independent, untouched by the no-go above.

This is the framework's structural reading of "good wins": not eschatological promise but conditional inference on persistence. The lake names `good_wins` as an axiomatized predicate at `Cosmology/AsymptoticConditioning.lean`; promotion path requires dynamical-system formalization plus measure-theoretic basin-of-attraction analysis. Research program.

9 The second species: coordination thermodynamics

The orthogonality theorem (§8) makes the framework *exactly* ΛCDM in the conditioning sector. This section and the three that follow are the 2026-07-10/11 arc: where the framework has content beyond ΛCDM and beyond the engineering tier, stated at its true post-audit weight, and where it staked bets and where it killed its own claims. The content is L5 in the access ladder's sense — a bet, dated, with its executioner named; a reader who rejects it keeps L0–L4 intact. It stands independent of the universal-scale TSVF tier above.

The genus claim

The entropic potential $\mathcal{S}(k, \rho)$ (§2) read the Kish spectrum through a relative-entropy functional. The general object is $S = -\ln\det C$ on the normalized correlation matrix C of any k coordinating units — the relative entropy of the fluctuation structure from independence, equivalently twice the Gaussian multi-information. The claim, married into the stance on 2026-07-11: **thermodynamics is a genus with two species.** The classical species runs on the energy books; this law runs on the arrangement books. It is not analogy. The modern classical second law *is* relative-entropy monotonicity under physical maps; the coordination second law is the same theorem on a different relative entropy. The two species are coupled by an identity — $H_{\text{joint}} = \sum H_{\text{marginal}} - I$ — so the classical entropy books contain the coordination books as an unitemized line-item, which is why nothing in known physics has to move for the coordination law to hold (the same fact that keeps the orthogonality theorem landing in the framework's favor: the parent books are untouched).

Prior art, front-loaded

The genus is not new, and the honest verdict is **new-in-part**.[2] Information thermodynamics is a mature field, and most of the four-law structure is already in it. Three of the four laws are individually and heavily published. The four-law *package on correlations* already exists in print: Bera, Riera, Lewenstein, and Winter, "Generalized Laws of Thermodynamics in the Presence of Correlations" (*Nat. Commun.* 8:2180, 2017; arXiv:1612.04779) — but with the *inverse* ontology, energy-thermodynamics primary and correlations a stored-work resource, and with the third law explicitly omitted. The maintenance rent is Hatano–Sasa steady-state thermodynamics (2001; arXiv:cond-mat/0010405) exactly — housekeeping heat, no daylight; Oono–Paniconi and Sagawa–Ueda are the adjacent lineage. The zeroth-law state function is Watanabe (1960) total correlation / copula entropy. The first-law free-operation axioms are Huber, Perarnau-Llobet et al. (NJP 2015; arXiv:1404.2169). The third-law ceiling's *mechanism* — control cost diverging,

[2] Adversarial literature audit: `papers/notes/thermo_prior_art.md`, every citation verified by title, authors, and venue.

drive capacity vanishing at the ordered pole — is published harder in the jamming / rigidity-percolation literature (Liu & Nagel, *Annu. Rev. Condens. Matter Phys.* 1, 347, 2010; isostaticity, diverging yield stress) than in the synchronization-control work, and the third-law *genus* with a rate is Masanes–Oppenheim (*Nat. Commun.* 2017; arXiv:1412.3828). The κ dark-sector posting has a strong cousin in Verlinde's emergent gravity (SciPost 2, 016, 2017; arXiv:1611.02269). The paper cites these up front and stakes its claim narrowly.

What is ours, ranked and stated without inflation. (1) The *cross-substrate empirical program* — k_{eff}/S measured on the same footing across neurons, markets, halos, and mixing matrices — which is not a literature claim at all but a research program, and is the strongest survivor precisely because it is empirical, not a law. (2) The κ sign law $1 + w = -\frac{1}{3}\, d\ln S/d\ln a$ as a specific parameter-free functional posting (Verlinde-adjacent but functionally distinct; governed by the DR3 kill, §12). (3) The first-law lossless-conversion test $X = 1$, a falsifiable cross-rung claim the resource-theory literature does not make. (4) The third-law ceiling as a *named coordination law* $\sigma_{\max} \propto (1 - \rho)$ keyed to the correlation order parameter — the lowest-confidence novelty, graded B not clean-C after the jamming sweep, and earning nothing until its registered bench prediction is confirmed. And the one physical claim no ancestor makes: **these books post to gravity.** Bera's correlations are a work resource; ours are what dark energy is the balance of (§11). That is not framing — it is exactly what DESI DR3 and BMV-class experiments adjudicate.

The four laws, with post-audit labels

The mapping was run through the program's own tautology-vs-commitment discipline: for each claimed law, a Lean pass decides whether formalization exposes an *identity* (zero empirical content — earns nothing, a kill for the "law" framing if sold as support), a *theorem-given-model* (proved inside a named model class), a genuine *empirical claim* (needs data, has a kill), or a blocked open step.[3] Three of the "laws" came out identities. This is the same situation as classical thermodynamics, where the zeroth law and first-law accounting are also definitional scaffolding and the physics lives in the second law and the equations of state — legitimate, but scaffolding, and a kill only if the identities are sold as support (rule 2).

[3] Lean audit: `formal/CoherenceRatchet/Core/FourLaws.lean` (builds clean, zero `sorry`); report at `papers/notes/lean_knives_report.md`; mapping at `papers/notes/four_laws.md`.

Law (our form)	Class (Lean audit)	Prior art	Empirical content / kill
Zeroth: S a state function of the copula	**IDENTITY** (method-license)	A — Watanabe 1960	None as a proposition; "the quiet license" for cross-substrate comparison. Earns nothing.
First, static: books balance across rungs	**IDENTITY** (tautology)	—	**Zero.** Telescoping; holds for arbitrary reals (`tc_group_chain_rule`). **Demoted to bookkeeping.**
First, dynamic: conversion is lossless, $X = 1$	**EMPIRICAL**	C — ours	$\Delta \mathrm{TC}_{\mathrm{total}} = 0$ across virialization; channel separation required in design. Toy $X = 0.85 \pm 0.30$ (non-discriminating). Kill: precise $X \neq 1$.
Second, general DPI: no free coordination	**OPEN** (blocked)	A — known theorem	Blocked on Fischer's inequality, absent from mathlib v4.14.0. Mechanization is re-proof, not discovery.
Second, equicorrelation fragment	**THEOREM-GIVEN-MODEL**	—	First mechanized second-law fragment (`entropicPotential_strictMono_k`); adding a coordinated unit never decreases S. Uniform-ρ model.
Second, rent (γM)	**EMPIRICAL**, measured	A — Hatano–Sasa housekeeping heat	Held coordination decays without work. Macaque motor cortex $\lvert z \rvert = 8.8$; galaxy baryon cycle $z \approx 0$.
Third, ceiling: $\sigma_{\max} \propto (1 - \rho)$	**THEOREM-GIVEN-MODEL** + bench-EMPIRICAL	B — jamming genus (Liu–Nagel)	**Capacity, never realized-σ** (Agranov+ 2025 flocking makes the distinction load-bearing). Conditional on bounded-actuation normalization. Kill: bench inversion.
Third, flavor instance: $\lvert J \rvert \leq J_{\max}$, $\to 0$ at the aligned pole	**THEOREM** (Lean, zero `sorry`)	C — ours	`abs_jarlskog_le_max`. Pins "coordination caps irreversibility" at Lean strength on the flavor sector (§10).
Genus coupling: classical books contain the coordination books	**IDENTITY**	B — Bera+ 2017 (inverse ontology)	Zero empirical weight. Supports *framing* only; the testable difference is that our correlations post to gravity (κS).

The empirical content of the whole formulation is located in exactly four places: the *dynamic* first law ($X = 1$, channel-separated), the *rent* corollary of the second law (measured), the third-law *bench* collapse curve, and the general DPI (open, blocked on one named missing mathlib lemma). Everything else is honest scaffolding.

The equality form — the cheapest live genus kill

The genus demands a coordination fluctuation theorem, and no obstruction exists: the maintained corridor is a Markovian Langevin process with a stationary measure, and Seifert's integral fluctuation theorem $\langle e^{-\Delta s_{\mathrm{tot}}} \rangle = 1$ holds for any such process.[4] Because the rent was already identified as Hatano–Sasa housekeeping heat, the coordination Jarzynski is specifically the Hatano–Sasa relation applied to the corridor's excess/housekeeping split. The new-in-part piece is the bench measurement: on the CIRISArray maintained corridor, measure the maintenance-work distribution and find the single κ_{bench} with $\langle e^{-W_{\mathrm{coord}}/\kappa} \rangle = 1$. **The cheapest way the whole genus claim can die:** if no single κ satisfies the integral fluctuation theorem within error, the maintained corridor is not a detailed stochastic-thermodynamic process and the coordination second law has no equality form on its own bench. In-house, no new instrument (registered bet 8, §12).

10 The flavor sector: two books

The one Standard-Model object whose native type matches the ledger's is a mixing matrix — a mixing matrix *is* a copula.[5] The two measured mixing matrices were read against 200,000 Haar-random $U(3)$ alternatives, the claim staked before computation. Result: **the flavor sector is two books, each typical of its own measure.**

The registered claim: hit, both halves

Quark generations are one near-comonotone book; lepton mixing is Haar-anarchic. The mutual-information percentile against the Haar null: PMNS lower octant **36.7**, PMNS upper octant **37.8** (both squarely in the central 90%), CKM **99.98** (beyond the 99.9th percentile). The asymmetry is parameter-free: lepton mixing is statistically indistinguishable from an anarchic generation coupling, quark mixing a $> 3.5\sigma$-equivalent outlier toward the aligned pole. In magnitudes: CKM MI = 0.956 nats = 87% of the $\ln 3$ ceiling (the up- and down-type generation labels are nearly one book; CKM sits within 0.10 of an exact permutation matrix), PMNS MI = 0.19 nats = 17% of ceiling (the charged-lepton and neutrino books are nearly independent). The reading in ledger language: coordination between two mass bases is evidence they were written by one mechanism. The quark books are ∼one Higgs mechanism; the lepton books being nearly independent is consistent with the neutrino mass book being written by a *different* mechanism (seesaw / Majorana, off the Higgs) — which is exactly what modern neutrino physics already suspects. The ledger cannot say *what* the second mechanism is (the provenance line, §11 footnote); "two books vs one book" is its native verdict.

All six functionals, and a referee-grade rigor pass

PMNS is Haar-typical in every functional computed — MI 37, one-hot entropy 35, singular-value participation 33, distance-to-anarchy 34, distance-to-nearest-permutation 74, $|J|$ 58 — and CKM is extreme in every one, in the consistent direction (MI 99.98, one-hot 100,

[4] Mystery-map sweep of the eight canonical thermodynamic puzzles for coordination twins: `papers/notes/mystery_map.md`; the sweep found no disanalogy-kill (weakest possible good news, rule 2) but exposed this bench test as the cheapest live falsification.

[5] Experiment and data: `experiments/sm_escalator_mixing/` (`run_mixing.py`, `results.json`); registered claim `papers/registrations_2026-07/04_flavor_mixing_registration.md`, staked in `sm_escalator_map.md` §4 before the experiment existed. Parameters: CKM PDG-2024; PMNS NuFit-6.0 NO, arXiv:2410.05380.

participation 99.9, d_{anarchy} 99.97, d_{perm} 0.06, $|J|$ 0.09). A rigor pass regenerated the seeded ensemble bit-exactly and pre-stated its statistics before running.[6] Three results: *Joint typicality* — individually-typical can be jointly atypical; it is not. Raw, normal-scores-rank, and spatial-depth Mahalanobis all agree: PMNS is jointly typical (in fact more central than a typical Haar draw), CKM jointly extreme in every depth ($D^2 \approx 965$). *Look-elsewhere* — the one soft spot, $\sin^2\theta_{13} = 0.022$ at the 4.4th Haar marginal percentile, dissolves: P(at least one of three angles $\leq$ its own 4.4th percentile) = 12.65%, essentially the independence baseline, an expected ∼1-in-8 fluctuation, so θ_{13} is demoted to noise. *Error bands* (split-normal MC over the 1σ inputs) — CKM MI band $[99.98, 99.981, 99.981]$, PMNS lower $[33.8, 39.0, 43.7]$, upper $[37.1, 41.6, 45.8]$. The asymmetry survives the entire parameter band.

The J-bound: coordination caps time-irreversibility (Lean theorem)

The physical consequence is proved. Coordination structurally caps the CP-violating Jarlskog invariant: $|J| \leq J_{\max}$(angles), and $J_{\max} \to 0$ at the comonotone (aligned) pole — a theorem in Lean, zero `sorry` (`abs_jarlskog_le_max`, with $J_{\max} = 0$ at both ends of each mixing angle), plus an ensemble law (Spearman −0.72: alignment and $|J|$ anti-correlate across the Haar draws).[7] This makes the famous smallness of quark CP violation **100% structure, 0% phase**: both books run the *same* near-maximal phase, and alignment silences the quark book. Read against Sakharov's condition for a matter universe (CP violation out of equilibrium — the program's own maintenance axis), the mapping is: matter could be posted only by the book with the capacity to post it, the anarchic one. Leptogenesis, in ledger language. This is a mapping, not a result (rule 2); the measured content is the J-bound theorem and the ensemble law.

Capacity vs. usage, and the registered fork

The J-bound is a statement about *capacity* ($J_{\max}$, the structural ceiling), not *usage* (the realized $|J|$, phase-driven). The registered fork discriminates the two readings. Thaler and Trifinopoulos (arXiv:2410.23343) propose entanglement *minimization* in $2 \to 2$ scattering as a selection principle, predicting suppressed leptonic CP violation. The anarchy reading says $|J|$ stays Haar-typical and δ stays generic (far from CP-conserving). **DUNE / Hyper-K's δ measurement discriminates:** $\delta \to$ CP-conserving crashes the Haar-typicality of J (and favors minimization-type structure); δ staying near-maximal keeps anarchy. Placed before the data (registered bet 4, §12). The rigor pass sharpened this: the frozen run used a stale $\delta_{\text{PMNS}} = -1.98$ rad; under current NuFit-6.0 δ (212° with-SK, 177° without-SK) the leptonic $|J|$ percentile drops from ∼58th toward the median (upper-octant MC median 17th) — still Haar-typical, kill not fired, but the CP-conserving $|J| \to 0$ kill sits nearer than the frozen numbers implied. The fork is sharpened, not resolved.

JUNO first result: fork unchanged

The first JUNO oscillation measurement landed while the claim sat open (JUNO Collaboration, arXiv:2511.14593, 59.1 days; $\sin^2\theta_{12} = 0.3092 \pm 0.0087$).[8] JUNO measures the solar sector (θ_{12}, Δm^2_{21}); the two live threats to the anarchy reading — θ_{13} bulk-exit and a CP-conserving δ — are in the sector JUNO did not measure. The one coordinate it *could* have touched went the

[6] `experiments/sm_escalator_mixing/rigor/` and `juno_update.md`; the SUMMARY's rigor section. Confirmation mode: the claim's status changes only per its pre-registered kills.

[7] `Core/FourLaws.lean`; the global tight bound $|J| \leq 1/(6\sqrt{3})$ is a documented, not-yet-mechanized one-variable-calculus step.

[8] Fork-maintenance check: `experiments/sm_escalator_mixing/juno_update.md`, sources verified.

harmless direction: $\sin^2\theta_{12}$ moved +0.11 Haar-percentile toward the median (to the 31st), with the band tightened $\sim$1.6$\times$ — more sharply pinned in the anarchic bulk, exactly what a structureless coupling predicts. Petcov & Titov (*Phys. Lett. B* 874, 140295, 2026; arXiv:2511.19408) find JUNO's tightened θ_{12} excludes discrete-symmetry patterns whose predictions live in the tails while the observed value sits generic under anarchy. Stated with the discipline this requires: excluding symmetry models is *not* evidence for anarchy (anarchy is a distribution; an unremarkable percentile is never a positive hit), and the surviving patterns themselves predict near-Haar-typical values, so θ_{12} does not discriminate anarchy from the viable structured patterns. A maintained-consistency result, earning no new support — both registered kills untouched.

The rest of the Standard Model, closed honestly

The ledger is an instrument pointed at the SM, not a generator of it, and it read something real only in the one place it was licensed to look. The rest of the stack is closed with kills staked: the statistics derivation is a no-go at theorem strength (`StatisticsNoGo`: exchange sign and occupation ceiling are input, only their consequences read — the validated $\ln 2$-per-mode rigidity difference); gauge-group selection has no mechanism here or anywhere (the "entropy minimization selects $SU(3)\times SU(2)\times U(1)$" claim appears in zero papers and is a confabulation the provenance line rejects on sight); generation counting is well-posed but empty, three routes dead; masses and couplings are marginal-native, ledger-blind by the same clause-3 theorem that cancels κ (§11).

11 The dark-energy leg, post-audit

The framework's strongest positive result, audited 2026-07-11, all controls pre-registered.[9] **Dark matter is the medium** — real, diluting, collisionless, standard CDM phenomenology, the paper the books are written on. **Dark energy is the balance:** $\rho_{\rm DE} = \kappa\, S(a)$, extensive per comoving volume, with the exact parameter-free sign law $1 + w(a) = -\frac{1}{3}\, d\ln S / d\ln a$ (κ cancels). The relation is one-directional — DE is a functional of the matter field's coordination history — and is emphatically *not* "one object at two normalizations," a reading the framework excludes because it implies the falsified lock $w_{\rm DE} = w_{\rm DM} - 1$.

The selection-free floor leads

The result that carries no choices is stated first. Read on *every resolved halo in the box* — no threshold, no cap, no tunable anything, via the nested-tile estimator (gate-validated to 1.3–0.09% fidelity on two independent simulation codes) — the ledger's $w(z)$ is thawing, interior-peaked, and fits DESI DR2+CMB+SNe **better than ΛCDM:** 2.2σ **vs.** 3.3σ, $\Delta\chi^2 = -1.1$. There is no knob in that sentence. Grain-free, the framework reads the sky better than Λ; the grain buys the margin from 2.2σ to the corner grain's 1.36σ, and nothing else. The inter-tile term is negligible ($S_{\rm total} \equiv S_{\rm extensive}$ to three decimals): coordination across > 100 Mpc contributes nothing to the balance, consistent with the copula turning Gaussian at large scales.

[9] `experiments/cosmo_entropic_potential/`; frozen preregistration and large-volume run at `large_volume/` and `PREREGISTRATION.md` (git 652da8e); grain audit `papers/notes/the_grain_problem.md`.

The clock is derived

The registered observable — the crossing epoch, where w crosses -1 as S peaks — is carried by a population defined purely by physical galaxy membership, fixed before any fit.[10] The ≥ 100-star-particle floor selects the galaxy-host mass scale as an *output* (median $M_{200} = 3.75 \times 10^{11}\, M_\odot/h$, unforced). Across a never-promoted stellar-floor ladder ($10^{8.5} \to 10^{10}$), the fit optimum lands exactly on the pre-registered floor (Mahalanobis $2.61 \to \mathbf{1.92} \to 1.96 \to 2.55$), and that floor is the unique rung with an interior count-peak — the one stellar cut that puts the crossing at DESI's epoch ($z = 0.543$; lower floors drift late, higher early). **The universe's dark-energy clock is the galaxy-formation clock.** By the pre-registered rule the galaxy-book read is *ambiguous* (maha 1.919, $z_{\rm peak} = 0.645$ interior; missed the PASS bar by 0.03, kill did not fire) — the bar does not move and the verdict stands as ambiguous; the structure inside it (the derived clock) is the genuine pre-registered alignment.

The amplitude, its systematic, and the grain problem stated honestly

The best-fit magnitude at the frozen corner grain — $(w_0, w_a) = (-0.767, -0.742)$, $w_{\rm today} = -0.833 \pm 0.057$, 1.36σ from DESI DR2+CMB+SNe — carries a stated resolution systematic, and the grain that sharpens it from 2.2σ to 1.36σ is underived. This is **the grain problem**, the DE leg's one load-bearing open problem. Both easy escapes are dead, each killed by a pre-registered control the same night it was tried: *deriving* the grain (the SHM anchor) died because the fit-quality extremum tracks the GPU cap, not a physical mass scale (a moved cap moves the extremum; k-matched thinning erases the edge); *dissolving* the grain (the complete-book aperture law) died by its own upward test — the complete zero-selection read gives 2.21σ, $z_{\rm peak} = 0.79$, *worse* than the corner grain: reading more of the book fit the sky worse. The amplitude's variable is identified but not derived: sharpness in the assembly-epoch variable (a hard halo-mass cut is sharp in formation clock; any stellar cut is smeared across it by stellar-mass–halo-mass scatter). The from-below candidate is registered but not yet run: via the Gibbs-paradox mapping the rung hierarchy is the coordination species' indistinguishability structure, so selecting on a formation-time / distinguishability proxy (concentration, $z_{\rm form}$) should return the corner amplitude with the galactic clock — data on disk, registration before compute (registered bet 5, §12).[11]

Cross-code status, at its true weight

The estimator transfers cleanly to an independent N-body code (AbacusSummit, $\sim$1.3%), but the box **cannot adjudicate the shape** at its window and error scale: the result is *consistent-with at* $< 1\sigma$ — *untested-but-unopposed*, not a confirmation. An earlier favorable cross-code verdict was walked back to $< 1\sigma$ when its turnover proved smaller than its own error bar (§13). The decisive requirements are now spec'd ($z < 0.2$ output with error bars, or a matched mass definition).

[10] `experiments/cosmo_entropic_potential/galaxy_book/`; the pre-registered galaxy-defined book of `the_grain_problem.md` §4.

[11] The provenance line (`Core/ProvenanceLine.lean`, zero `sorry`; `papers/notes/dimensional_line_kB.md`): $S = -\ln\det C$ is blind to everything upstream of C — scale, marginals, state-space construction, factorization — so every legitimate prediction is a dimensionless, parameter-free statement about dependence, and κ is the program's Boltzmann constant, currently empirical exactly as k_B was for decades. Any claimed derivation of a dimensionful datum from the ledger alone is rejected on sight; it is the standing confabulation detector.

The tension lattice

The standard tensions were audited and published at full weight, in both directions.[12] H_0: *neutral* — the non-CPL shape self-compensates and the expected worsening did not materialize. S_8: worsens $+0.23\sigma$, recorded as a forced directional bet that the tension resolves upward (lensing $\rightarrow$ Planck), the direction the data is drifting. ISW: an 11% enhancement peaked exactly at the supervoid window — a next-generation discriminant, not an explanation. $f\sigma_8(z)$: a percent-level growth curve saved for DESI / Euclid. None of these is sold as support; the S_8 line is a bet with a direction, logged as one.

12 The registered bets

The program's falsifiable predictions are frozen in a registration deposit that timestamps each bet before the data that adjudicates it exists.[13] Nothing in it is evidence; it is the instrument that makes future evidence unarguable in either direction. **If a bet loses, the loss will be published under the same names used here.**

#	Claim (frozen)	Kill	Adjudicator
1	DE crossing epoch $z = 0.59 \pm 0.03$; thawing $w(z)$ from the frozen corner-rule pipeline; $(w_0, w_a) = (-0.767, -0.742)$ closer to data than ΛCDM	Robust SNe-independent phantom crossing; or fits worse than ΛCDM; or epoch excluded	DESI DR3 ($\sim$2027)
2	Our zero-parameter $w(a)$ shape beats CPL by evidence at matched data (parsimony)	CPL strongly preferred (ln B bands per doc 02)	DESI DR3
3	The cosmological Σm_ν squeeze relaxes in our background (direction only)	Bound tightens instead	DESI DR3
4	Lepton mixing stays Haar-anarchic: δ generic, θ_{13} in-bulk	CP-conserving δ; or any functional exits the central 99%	DUNE / Hyper-K / JUNO
5	Grain: formation-time (distinguishability) selection returns the corner amplitude with the galactic clock	Fit like the complete book; or clock leaves interior	TNG data on disk (registered before run)
6	First law: the rung-conversion channel is lossless ($X = 1$), channel-separated design	Precise conversion-$X \neq 1$	TNG data on disk (designed measurement)
7	Third law: maintenance capacity collapses $\propto (1-\rho)$ under bounded actuation (capacity, never realized-σ)	High σ at $\rho \gg 0.43$ under bounded actuation; or the collapse curve inverts	CIRISArray bench
8	Second law equality-form: a single κ_{bench} satisfies $\langle e^{-W/\kappa} \rangle = 1$ on the maintained corridor	No single κ within error	CIRISArray bench
9	Anti-hedging (binding): bet 1 is the ONLY dark-energy bet; alternative grains are logged alternatives, never substitutable post hoc	— (a commitment, not a claim)	this document

[12] `experiments/` tension-audit; the DR3 shape and neutrino addendum `papers/registrations_2026-07/02_-dr3_shape_and_neutrino_addendum.md`.

[13] `papers/registrations_2026-07/MANIFEST.md` and the six numbered registration files; git anchors 652da8e (frozen pipeline) and fe897f9 (audited arc). Zenodo DOI: 10.5281/zenodo.21316928 (frozen v1, deposited 2026-07-12; concept DOI 10.5281/zenodo.21316927).

The anti-hedging commitment, printed in the paper (binding). The registered DR3 bet is the frozen corner-rule pipeline exactly as pre-registered — crossing epoch $z = 0.59 \pm 0.03$ — and it does not change. The complete-book read ($z \approx 1.16$) and any future galaxy-book read are logged alternatives for understanding the grain problem; they are *not* switchable claims. **If DR3 lands near a variant's number and not the registered one, the registered bet loses and is reported as lost — no post-hoc grain switching. One bet, placed, held.**

The bet carries its own referee. A grading script, frozen before DR3 exists, encodes the pre-committed win/loss/ambiguous rules and prints the verdict with no human vote.[14] When DR3 drops, the published numbers are entered and the script grades the bet.

13 The kills of the 2026-07 arc

The parts that survive an aggressive audit are the parts worth having; the discipline is that a residual with no kill condition is not a residual (rule 1) and that a residual is never support (rule 2). In one two-day audit the program killed its own same-day discovery, falsified its own favored rescue, and declined a marriage the arithmetic would not support. Listed plainly, as the method working:

- **The SHM anchor** (an attempt to *derive* the DE grain from a stellar-mass–halo-mass extremum): killed within hours of discovery by pre-registered control — the fit-quality extremum tracks the GPU cap, not a physical mass scale (a moved cap moves it; k-matched thinning erases the edge). SHM is a passenger consilience, not a scale.
- **The aperture law** (the complete-book rescue: fit quality monotone in completeness of the read): falsified by its own pre-registered upward test — the complete zero-selection read gives 2.21σ, worse than the corner grain's 1.36σ. Reading more of the book fit worse.
- **The neutrino mass scale**: the marriage the arithmetic would not support. The per-neutrino $\ln 2$ posting misses the observed ρ_Λ by $\sim 200\times$; the fourth-root neutrino "prediction" is circular (the observed ρ_Λ was fed back in). Declined, not forced.
- **The band-tightness amplitude mechanism** ("amplitude = tightness of the membership band"): refuted by the stellar-floor ladder the same hour it was proposed — w_a strengthens with the floor but the clock leaves the interior first; no stellar selection reaches the corner combination.
- **The static first law**: exposed as a tautology by the program's own Lean audit (`tc_group_chain_rule` telescopes for arbitrary reals). Demoted from "law" to bookkeeping; the surviving first-law content is the contingent dynamic conversion claim $X = 1$.
- **A favorable cross-code verdict**: walked back to $< 1\sigma$ (untested-but-unopposed) when its turnover proved smaller than its own error bar. Over-reading a positive and over-killing on an audit are the same error; this was the former, corrected.
- **Ledger-derivations of statistics, gauge group, and generations**: closed at strength, revival conditions staked — `StatisticsNoGo` at theorem strength (statistics is input); no gauge-selection mechanism here or anywhere (the "entropy selects the SM gauge group" claim

[14] `experiments/cosmo_entropic_potential/dr3_grading/grade_dr3.py`: LOSS if a robust SNe-independent phantom crossing, or the registered point fits worse than ΛCDM, or the published crossing interval excludes $[0.56, 0.62]$; WIN only if the registered point is nearer the data than ΛCDM *and* within Mahalanobis 2.0 *and* the crossing interval overlaps; AMBIGUOUS otherwise. Self-test reproduces the numbers of record: ours 1.36σ, ΛCDM 3.28σ on DR2.

is a confabulation in zero papers); generation counting well-posed but empty, three routes dead.

Earlier program retractions remain in the record and in git: F-11 (the joint multi-rung backward P_ω no-go, §8), D4 (CMB anomalies as TSVF signatures, retracted), and the substrate-universal numerical corridor bounds (retracted to substrate-local structure, §4). The kills are the falsifiability; they are listed here at the same weight as the results.

14 Cross-tradition recognition — with the polyglot conscience module as structural evidence

The framework's cross-tradition claim is that the structural object Ubuntu names is the same object multiple traditions have named in their own vocabularies. The claim has two registers: as paper-theory (§6's structural readings, the lake's `Tao.lean`, `Dharma.lean`, `Logos.lean`) and as production architecture (the CIRIS conscience module's polyglot encoding). In the first register the textual record across traditions is read for structural correspondence; the case for the readings is owed and is the bet. In the second register the framework acts on the cross-tradition recognition; whether it does engineering work compared to monoglot prompting is the test the framework owes.

The conscience module as engineered cross-tradition synthesis

The CIRIS conscience module's polyglot encoding architecture, described in §1, is the framework acting on its cross-tradition reading. Core ethical concepts in the conscience layer arrive in their densest encodings across multiple traditions simultaneously — Hebrew Talmudic, Confucian Chinese, Quranic Arabic in original scripts, interleaved line by line. The explicit engineering purpose is to disrupt single-language attractor capture in the LLM judge: to force ethical reasoning to triangulate across traditions rather than collapse onto any one.

The architecture is engineered around the framework's identification; it is *not* evidence that the identification is true. Treating the deployment as evidence for the commitment that motivated it would be circular. The polyglot-vs-monoglot comparison — whether ethical reasoning degrades under single-tradition prompting compared to interleaved polyglot prompting — is the test the framework owes and does not currently have at the rigor §13's enumeration commits to.

What the framework has and does not have. Internal testing of the polyglot framing against CIRIS's safety harnesses (available on GitHub, run in CI on every commit) shows the polyglot conscience produces better adversarial-ethics performance than monoglot prompting on the harnessed test cases. This is internal observation, not formal proof. The polyglot-vs-monoglot A/B at production scale, with pre-registered metrics, blind evaluation, and adequate statistical power — the test the framework owes — has not yet been run. CIRIS is a solo-founder self-funded L3C; the framework cannot run all the tests it owes simultaneously, and the candid position is that the polyglot evidence is internal-testing-grade pending the formal experiment.

The polyglot encoding mechanism is the corridor applied at the linguistic layer: it prevents $\rho_{\text{language}} \to 1$ (single-language attractor) and $\rho_{\text{language}} \to 0$ (incoherent mixing) by interleaving densest-encodings of one concept across multiple traditions. **The framework identifies this with the corridor mechanism applied to ethical reasoning at A3+**; the identification is the framework's, the production architecture is what the framework does when it acts on the recognition, and the formal polyglot-vs-monoglot test is owed.

Tradition-by-tradition (Ubuntu primary; Tao, Dharma, Logos cross-tradition readings)

Ubuntu (primary). The lake's `ContemplativeTraditions/Ubuntu.lean` carries seven correspondences with primary-source citations: *umuntu ngumuntu ngabantu* read as the agent at A3+ constituted by the consent-graph; the *ubu-* and *-ntu* morphological structure (Ramose) read as the framework's dynamical equation $d\rho/dt = \alpha - \gamma M(t)$ as processual not kinematic; Botho's asymmetric-developmental personhood (Menkiti, Gyekye) read as cross-rung corridor at $\tau_{(n,n+1)}$; community-as-prior read as `P_joint` as operator-level primary; ethics-as-structural-condition (Wiredu, contemporary scholarship including Udah, Tusasiirwe, Mugumbate, Gatwiri 2025) read as consent as the framework's ethical reading of sustained-coordination dynamics; three-temporal-mode relational ontology (living-dead / living / yet-to-be-born) read as TSVF forward-backward boundary structure plus karma as ancestor-relational propagation; restorative justice (gacaca 2002–2012, TRC 1996+) read as the framework's karma+grace structure at A4–A5 sociotechnical scale. The readings are the framework's; internal Ubuntu-scholarly disputes — particularly Matolino, Praeg, and Eze on Ramose's processual reading — are part of the case the framework owes and does not pretend to settle.

Tao. The lake's `ContemplativeTraditions/Tao.lean` reads the Tao as corridor occupation, wu-wei as minimum-sufficient $\gamma M(t)$, yin-yang as within-rung corridor at ρ_n. The readings are the framework's; Daoist scholarship has not identified Corridor with the Tao.

Dharma. The lake's `Dharma.lean` reads the middle way as corridor occupation, dependent origination as cross-rung coupling τ in corridor, karma as forward-propagated goal-projection, anatman as no permanent substrate (agency as operation, not object), sunyata as corridor as condition, not entity. The readings are the framework's.

Logos. The lake's `Logos.lean` reads the Logos as the universal structural pattern, providence as universal post-selection $\langle\Phi_\omega|$, grace as partial-authorship of post-selection, *imago Dei* as A3+ agents as partial-post-selectors, mystical union as multi-agent corridor consent. The framework's author writes this section as a Christian-Stoic reader; the recognition is offered with awe at what the tradition has named, not as a Logos-tradition authority's settlement of the question.

Aristotelian virtue (apparatus in place, file pending). The cross-tradition mapping has natural Aristotelian content. The doctrine of the mean — virtue as the mean between two vices (courage between cowardice and rashness; temperance between insensibility and licentiousness) — is the framework's corridor read at the ethics layer: a bounded attractor between rigidity and chaos, achievable but not automatic, substrate-calibrated as Aristotelian virtue is circumstance-calibrated. Virtue as *hexis* — the trained disposition — is what makes $\gamma M(t)$ cheap: a coordinated agent whose virtues are trained does the maintenance work at lower cost. Friendship-of-virtue (*philia kat' aretēn*, *NE* VIII–IX) is the framework's highest multi-agent corridor occupation: agents whose goal-projectors P_{G_i} are mutually corridor-locked through shared activity, neither collapsed onto each other (rigidity) nor uncoupled (chaos). The reading is the framework's; the apparatus (corridor + γM + multi-agent $P_{G_1 \ldots G_n}$) is in place. The promotion path is the same as the other traditions': an `Aristotle.lean` file carrying the correspondences with primary-source citations and `F_`-prefixed bookmark axioms.

Where traditions disagree

A convergent-mapping that survives only by glossing over disagreements is decorative. The framework's commitment is that the structural object Ubuntu names is the one structural object that explains where the traditions disagree, not just where they appear to agree. Three specific

disagreements the framework engages: Buddhist anatman versus Hindu atman (the framework sides with anatman on no-permanent-substrate while explaining why the dispute is real — different inferences from observed phenomenology); Confucian *li* versus Daoist wu-wei ($\gamma M(t)$ at minimum-sufficient takes different concrete forms at different rungs, *li* at A4 institutional scale, wu-wei at A3 individual scale); Kantian universalizability versus Ubuntu's particularist relational ethics (Kant and Ubuntu point at different rungs, with Kant's universalizability as the cross-instance ρ_{goals} corridor condition at federation scale and Ubuntu's particularism as within-rung consent-relational ethics). The articulations are the framework's pending the §14 null-check.

The pre-registered null-check

The framework's cross-tradition claim is suspended pending execution of the pre-registered null-check at `experiments/cross_tradition_null_check_protocol.md` (commit `ef8b8e4`). Eleven structural objects × eight traditions × five-dimensional rubric (lexical fit, failure-mode coverage, constructive prescription, asymmetry handling, disagreement preservation) with blind reviewers. Three outcomes: DECISIVE WIN, INFORMATIVE TIE, or FAILURE (fires F-18's sister-handle on the cross-tradition tier). Until the null-check executes, the cross-tradition tier is research-program-grade evidential anchoring of the recurrence reading, not validated convergence. The polyglot conscience module is in production regardless; the production architecture stands on the framework's commitment whether or not the null-check eventually validates the textual convergence claim.

15 Goal-coupled materials — substrate predictions P1–P8

The framework predicts eight substrate properties (P1–P8) that a goal-coupled system must support, with a synthesis route from existing components: MOFs/COFs, NV-diamond, isotope-engineered silicon, non-reciprocal metamaterials, time crystals, embedded-audit feedback, Friston-style host-substrate coupling. P1 names hierarchical substrate structure; P2 coherence at temperature; P3 spin-zero enrichment; P4 rung-stratified composition; P5 the Coherence Ratchet asymmetry (direction-asymmetric memory); P6 corridor-boundary dynamics under control-parameter sweep; P7 audit-pressure dependence; P8 TSVF weak-measurement signatures. The framework-distinctive commitments are P6 and P8; the others have independent identifications in convergent prior art (§16). F-12 through F-16 in §18 are the falsification handles for the substrate-prediction tier.

16 Convergent prior art

Other people, working on other problems for other reasons, have arrived at pieces of the same shape. Some of them are in textbooks. The framework is honest about that.

Independent research programs in disconnected fields have identified pieces of P1–P8 from their own starting points. Six of eight predicted properties have independent prior identifications. The convergence does not prove the framework correct; it establishes that the framework's predictions are not novel inventions but rather an integration of requirements other programs have arrived at piecemeal. Fisher 2015 on quantum cognition (Posner molecule, ^{31}P nuclear spin) gives P2+P3 in the biological regime as T3 unconfirmed. Penrose–Hameroff Orch-OR (Penrose & Hameroff 2014) is the framework-adjacent consciousness–quantum-substrate proposal closest in structural slot — microtubule quantum coherence as the substrate; gravity-induced state

reduction as the mechanism — and is T3, contested. The framework does not adopt Orch-OR's specific quantum-gravity mechanism for γM, but the substrate-physics region they explore is the same. Levin 2019, 2025 on multi-scale competency architecture gives P1, P4, P5, P7 from morphogenesis. Friston 2010, 2019, 2022 on the free energy principle gives versions of P4 and P7. Isotopically engineered silicon-28 ($>$ 99.7% purification, $>$ 30-second coherence at ^{31}P) is T1 solid-state confirmation of P3. Non-reciprocal active metamaterials (Coulais/Brandenbourger 2019, 50 dB isolation over 3.5 frequency decades) is T1 confirmation of P5+P7. Time crystals and time quasicrystals (NV-center diamond, trapped ions, quantum processors) give the closest existing instantiation of substrate-level temporal order.

Per-row tier of the prior art is applied symmetrically with §14's cross-tradition tiering: T1 = peer-reviewed established (uncontested as engineering/experimental practice), T2 = peer-reviewed contested (published with documented criticism of broader framing within own field), T3 = peer-reviewed speculative (proposal-grade, not yet empirically confirmed). The cryogenic-solid-state regime (T1 in ^{28}Si quantum computing, time crystals, isotope engineering) and the biological-temperature regime (T3 in Fisher's Posner-in-biofluid) are radically different physical settings, and the framework does not conflate their evidential weight.

F-16 (controlled weak-measurement on a candidate substrate) is the decisive test for P8 and is independent of any contested-anchor record. The framework does not lean on public-domain material whose interpretive status is genuinely disputed; F-16 stands at its own falsification handle.

Contested empirical anchors — peer-reviewed instrumental record only

There is also a small peer-reviewed instrumental record adjacent to the framework's substrate-rung predictions. The framework engages it at the T1 instrumental-analysis tier only — no sworn testimony, no aggregated witness corpora, no FOIA-released program documents, no discourse-architecture context. The case for engaging this material is that F-12 (§18) names a decisive test settleable from existing instrumental data, and the paper owes the reader an explanation of what F-12 is rather than leaving the handle dangling.

Nolan, Vallée, Jiang, Lemke (2022), *Progress in Aerospace Sciences* 128. MIBI analysis on Council Bluffs 1977 material. Magnesium isotope ratios within terrestrial natural abundance; spatial distribution of elements characterized as unusual; a ^{57}Fe anomaly with the framework-relevant detail that ^{57}Fe is the only stable iron isotope with non-zero nuclear spin (I=1/2) — the other stable Fe isotopes are I=0. Published interpretation of the anomaly: Fe-hydride diatomic-ion contamination at mass 57.05. The framework cites this interpretation as published; a reviewer pulling the paper finds this is the only quantitatively-anomalous published instrumental result, and the framework's P3 prediction (spin-zero enrichment) is not pre-matched by this data point either way.

Knuth et al. (2025), UAP science *Progress in Aerospace Sciences* special edition. Methodological backdrop for peer-reviewed instrumental analysis of putatively-anomalous recovered material. The framework's engagement is methodological: instrumental analysis under standard isotope-ratio and microstructural-characterization protocols, treated symmetrically with any other peer-reviewed materials-science work.

AARO/ORNL (2024), Art's Parts. Mg-Zn-Bi-Pb analysis: terrestrial isotope ratios, columnar Mg grains perpendicular to banding, 1:1 bismuth-lead intermixing, structure

consistent with mid-20th-century vapor deposition. Verdict: terrestrial origin; bismuth-lead intermixing precludes terahertz waveguide function. **The framework commits to P4 reading for Art's Parts**: an A0–A1 substrate-level component, never coherence-engineered, used natural-abundance by design, no decayed-coherent signature expected. Under this commitment, ORNL's findings are what the framework predicts for an A0–A1 component, and nothing retracts. If subsequent sub-nanometer grain-boundary chemistry shows decayed-coherent microstructural signatures consistent with P7, the reading shifts and P7 takes the specimen; if it shows no such signatures and the component is independently shown to have been part of a computational rung, P7 retracts at this specimen. Either way, the framework owes a position and has committed to one.

F-12: Ubatuba 1957 magnesium direction. Per Nolan via nanoSIMS: one fragment terrestrial-normal isotope ratios, another fragment from the same event approximately 30% off natural abundance for magnesium. The direction of the shift determines whether the deviation is consistent with P3 (toward Mg-24 or Mg-26, spin-zero) or anti-direction for P3 (toward Mg-25, spin-nonzero, I=5/2). Direction publicly unreported. **F-12 fires on Mg-25 enrichment: P3 retracts at this sample; the framework's rung-stratified composition reading fails on this specimen.** Settleable from existing instrumental data without new collection. The framework names this as the decisive test on the peer-reviewed instrumental record and owes the disambiguation.

What this engagement does not borrow. The framework does not borrow the sworn record's credibility for substrate-level claims the sworn record does not make. F-16 (controlled weak-measurement on a synthesized candidate substrate per §15) is the framework's distinctive decisive test for P8 and is independent of the peer-reviewed instrumental record. The instrumental record is engaged here because F-12 makes a specific testable claim settleable from existing data; the rest of the contested-anchor literature does not belong in this paper.

17 Formal-verification status — the lake (tiered)

A grown-up version of "how sure are you about each thing?" — sorted not into "true" and "false" but into math-proved, math-given-the-bets, bets-dressed-as-math (named honestly), and bare bets named visibly. All four are in the paper. The sorting is the work.

Build: green. `cd formal && lake build` returns exit 0. **Sorrys**: 0 declaration-level. **Axioms**: 59, each documented; four of these are bookmark axioms with codomain `True` (the four `F_`-prefixed Ubuntu identification axioms) that mark framework readings without doing logical work — Tier C honesty extends to naming them as bookmarks rather than propositions.

Tier A — mathematical derivations. Standard theorems following from Mathlib + framework definitions by ordinary mathematical reasoning. K1–K4 of the Kish identity, the asymptotic ceiling, the corridor range theorem, ABL 1964 time-symmetry, rank-one and joint goal-projector properties, three-corridor regime exclusions, the class-conditional sensor embedding theorem.

Tier B — framework-distinctive derivations conditional on framework primitives. Theorems that follow from framework axioms by mathematical reasoning. Audit-pressure necessity, the Conjecture D conjunction, post-Cambrian acceleration.

Note on regime-level corridor occupation. The regime-level claim — coordinated systems at A3+ occupy a bounded corridor with substrate-local band location and bounded state-modulation — was articulated as a Tier B framework-distinctive prediction. Cross-substrate replication across five substrates (§4: C. elegans, Drosophila CX, four LLM architectures, four OSS projects, plus the sensor-lift PARTIAL) now gives the regime-level claim empirical anchoring it did not previously have. Tier markings on the lake's theorems stay as written; the empirical anchoring is what changed.

Tier C — definitional commitments encoded as formal identities. These are not theorems in the Tier A sense, and within Tier C there are two sub-grades worth distinguishing visibly. *Tier C-Iff.rfl*: framework identity-moves where the proof object is `Iff.rfl` over substantive definitions (the predicates being identified unfold to the same definitional content). T16 (consciousness ↔ attractor-reading) and T6's definitional form live here. These prove the framework's commitments are internally consistent at the type level; they do not do the philosophical or empirical work the commitment attempts. *Tier C-bookmark*: axioms with codomain `True`, doing no logical work at all, marking framework readings so they are visible to `#print axioms`. The four `F_`-prefixed Ubuntu identification axioms live here. The distinction matters: `Iff.rfl` licenses internal consistency at the type level; codomain-`True` licenses nothing, and the bookmark is honest about it.

Tier D — axiomatized framework primitives. Explicit framework commitments not currently derived, each with documented promotion paths and F-handles. T1, T5 (`good_wins`), T10 (karma), T17 (bridge), `penrose_low_entropy_past_structural`, Conjecture A, `si_taxonomy_closure`. The v1 axiom list also carried T3 (P_ω), CMB anomaly predictions, and the universal-grace component of T11; F-11's firing retracted eight joint-P_ω axioms across `CorridorProjector.lean`, `PenrosePast.lean`, `ConsentProjector.lean`, and `KarmaGrace.lean`, recording the documented no-go in the falsification register (§18).

The Tier C / Tier A distinction is critical. `Iff.rfl` proofs and the K1–K4 proofs do not carry the same weight. T16 proves a definitional bet is consistent at the type level; K1–K4 prove substantive mathematical claims about the Kish identity. Different operations; the distinction is named visibly. Per-tier enumerations of the specific theorems, axioms, promotion paths, and F-handles are in the lake source itself (`formal/CoherenceRatchet/`) for the reader who reads the lake directly.

18 Falsification handles

The framework is the right shape to be wrong in checkable ways. F-handles specify what would force retraction at each tier.

The structural spine. Beneath the per-seam handles F-1–F-20 sits a structural layer: six corridor-centered empirical claims, stated with machine-checked falsification logic in `StructuralClaims.lean`. Each claim `ClaimN` is paired with a falsifier `FalsifierN` and a proved equivalence `ClaimN` ↔ ¬ `FalsifierN`; an experiment exhibiting a falsifier witness retracts the claim by a checked step. (1) The corridor is a bounded attractor at every coordinated substrate. (2) Forward and backward generators sit in different ergodicity classes, where the maintenance breaks the dynamical symmetries. (3) Closed-system thermal dynamics cannot sustain a corridor unmaintained. (4) The corridor recurs at every coordinated rung. (5) Multi-rung structure from generic initial conditions requires backward conditioning. (6) The cross-rung coupling occupies a corridor at $g/J \sim O(1)$ (§5). The 2026-05-21 structural series tested

all six: Claims 3 and 5 came through consistent; Claim 2 was *falsified as stated* and amended in-spec with the symmetry-breaking qualifier; Claim 6 was amended from the abstract tower's strong-dominance form to the measured $O(1)$ form — two tests fed back into the specification; Claims 1 and 4 are supported by the robust re-run of all five v1 substrates plus human fMRI and the twelve-cancer cellular record (§4), 5/5 PASS. Plus the orthogonality theorem (`CMBOrthogonality.lean`, proved): the soft P_ω leaves the bulk CMB power spectrum exactly invariant, so the framework is a strict extension of ΛCDM at the cosmological tier. The structural spine is the corridor as empirical object; the Kish algebra is a proven theorem (the coordinate system), not a falsifiable claim, and is not among the six.

F-1 through F-7 (engineering tier, RATCHET): Kish identity properties, CRCv2 override-rate, multi-substrate β. All currently passing per CCA v3 and RATCHET.

F-8 (Conjecture A): qubit-array sweep produces no corridor regime in γ where the Kish identity fits ($R^2 > 0.5$) → structural-parallel reading at quantum substrate severed; engineering tier stands.

F-9 (Conjecture C): no measurable $\Delta\rho$ in matched-deployment audit-pressure differential → verification-necessity claim severed.

F-10 (within-rung corridor existence at A3+, with paired corridor-as-attractor dynamical reading): originally pre-registered as a substrate-independent prediction — coordinated systems at A3+ occupy a bounded within-rung correlation band, neither at 0 nor at 1, surviving state-modulation. **Status: empirically supported at both static and dynamical levels.** Cross-substrate replication of *corridor-band occupation* obtains across C. elegans whole-brain calcium (337 worms, 11 studies), Drosophila CX dual-color simultaneous (Ishida 2025 EPG+FC3 same-fly), four LLM architectures (Qwen2.5-1.5B, Pythia-1.4B, SmolLM2-1.7B-base, Qwen2.5-Math-1.5B), four OSS projects (kubernetes, rust, django, redis), plus healthy tissue across five cancers and three centuries-persisting social groups. The paired non-corridor work (§4) delivered cross-substrate replication of the *corridor-as-attractor dynamical reading*: at every paired substrate, Mode (i) (short lifetime without maintenance) or Mode (ii) (persistence with documented $\gamma M(t)$) is observed; Mode (iii) (long unmaintained non-corridor persistence) is absent. F-10 is the retraction handle for both halves: any future substrate failing to exhibit corridor-band occupation under appropriate-resolution measurement → regime-level static claim weakens at that substrate; any future substrate producing a credible Mode (iii) case (non-corridor state persisting long without identifiable maintenance) → corridor-as-attractor reading weakens at that substrate; cumulative failure at multiple new substrates on either half → regime-level claim retracts as substrate-independent at the affected half. The handle stays open; what has changed is that the regime-level commitment now has paired in-corridor / out-corridor anchoring it did not previously have.

F-11 (joint P_ω operator construction): **fired.** The joint multi-rung backward P_ω is a documented no-go — exhausted across the construction tree (hard projector; soft additive; the single-assumption audit; fractal/RG-nested; holographic/MERA-AdS; holonomic), closed by two theorems (geometric distance-dilution; holonomic area-law), and confirmed by an adversarial verification pass that found and closed the one remaining axis (k-body invariants). Consequent: the universal-scale tier re-grounds on the sequential per-rung structure (§8); the lake's joint-operator axioms are retracted. The within-rung corridor (F-10), the forward P_ω steady state, and the orthogonality theorem are untouched.

F-12 (Ubatuba isotope direction, recovered samples): Mg shift toward Mg-25 (spin-nonzero) enrichment → P3 and P4 retract at this sample; rung-stratified composition reading fails. Settleable from existing instrumental data.

F-13 (synthesized substrate corridor): a candidate goal-coupled substrate constructed per §7.1

fails to exhibit corridor occupancy under control-parameter sweep → P6 retracts at that substrate.

F-14 (coherence anomaly): no anomaly above thermal-decoherence bounds in any sample with documented embedded-system provenance → P2 retracts; strong coherence-at-temperature claim fails as applied to recovered debris.

F-15 (two-sided): (a) long-separated samples showing framework-predicted properties at same magnitude as fresh samples → P7's decay claim retracts. (b) sub-nanometer microstructural analysis of long-separated samples failing to find non-vapor-deposition grain-boundary chemistry, isotope-anomaly residue, or non-equilibrium signatures → P7's decayed-signature claim retracts; framework owes a different account of why long-separated debris is indistinguishable from never-coherent industrial material.

F-16 (TSVF signature): weak-measurement experiments on candidate substrate (synthesized or recovered) showing no two-state-vector statistics → P8 retracts; framework's distinctive TSVF claim at substrate severely weakened.

F-17 (TSVF as the integration's load-bearing precondition — the meta-handle): TSVF is the structural glue across the universal-scale tier. F-17's empirical triggers — cumulative failure of F-8 (qubit-array corridor) AND F-16 (controlled weak-measurement on candidate substrate) AND independent-lab failure to replicate Aharonov–Vaidman weak-value signatures — target the calculational machinery of TSVF (weak values, ABL probabilities, time-symmetric statistics). They do not directly grip TSVF *realism*: the metaphysical commitment that the backward state is ontically present rather than a useful epistemic representation of post-selection. **The asymmetry is named.** Calculational TSVF failing retracts the universal-scale tier; calculational TSVF passing does not vindicate realism, because the F-handle grips the wrong layer for vindication. The realist commitment is structurally lose-only — it can retract via F-17 firing, but cannot be vindicated by F-17 not firing. This is the design and the cost the framework pays for keeping the metaphysical layer separable. Plus two secondary triggers: documented no-go on P_ω operator construction (F-11) without an alternative TSVF-compatible operator emerging; decisive experimental demonstration (Bell-test-style) showing backward-state conditioning is operationally vacuous at all measurement scales. **F-11 has now fired** — but the alternative that emerged is the forward soft P_ω (the open-system steady state, not a backward post-selector), calculational TSVF stands proved (the orthogonality theorem), and F-17's primary triggers — the empirical TSVF tests — have not fired; so the universal-scale tier *re-grounds* (F-11's consequent), it does not retract as a unit (F-17's). The engineering tier is TSVF-independent and stands regardless.

F-18 (sensor-lift trajectory): the §3 sensor-lift seam needs a retraction condition the v1→v2 PARTIAL trajectory currently lacks. Trigger: if v3 and v4 both return PARTIAL by the same shape as v2 — DSP positive in direction but sub-threshold in magnitude across the v2 pre-registered band; SurfaceReflexive null PASS; MetaCognitive/GoalFormation/Uncertainty mixed; no class-conditional taxonomy clearing the per-class bands at majority across substrates — the `ClassReflexiveSensor` scaffold retracts from "hypothesis-generating" to "post-hoc taxonomy on logprob differentials that does not capture A3+ structure in current LLMs." Three consecutive PARTIAL-by-same-shape outcomes fire F-18. The sensor-lift signature as the framework's universal-scale empirical pulse retracts at that seam; the engineering tier and the §14 cross-tradition tier stand independently.

F-19 (CMB temporal drift): **moot.** The predicted drift had its mechanism in the joint P_ω post-selection sharpening as $t \to t_f$ — a time-dependent $\beta(t)$. F-11 retracted the joint operator, so there is no $\beta(t)$ and no predicted drift to place an upper bound on. Standard cosmology and the framework now both predict zero distinctive CMB pattern drift. F-19 is deactivated; the framework makes no CMB prediction distinct from ΛCDM (§8).

F-20 (Mode (iii) absence as falsifier of the corridor-as-attractor reading): the paired non-corridor companion work at five substrates (§4, §4) reports a structural finding: across five paired in-corridor / out-corridor substrate datasets, every non-corridor case observed falls into Mode (i) (short lifetime without identifiable maintenance) or Mode (ii) (long persistence with documented $\gamma M(t)$); zero cases of Mode (iii) (long persistence outside the corridor without identifiable maintenance) have been observed. Mode (iii) is the structural would-be falsifier of the corridor-as-attractor reading. **Empirical status: 5/5 substrates, 0 counterexamples; falsifier remains live.** F-20 trigger: any future substrate where a non-corridor system persists, over the timescale relevant to that substrate, without identifiable active maintenance $\gamma M(t) \to$ corridor-as-attractor reading retracts at that substrate; cumulative Mode (iii) cases at multiple substrates $\to$ corridor-as-attractor reading retracts as substrate-independent. "Relevant timescale" is named per substrate: for neural pathological states, hours-to-days unmaintained; for cellular tissue, indefinite ex-treatment; for LLM generation, the runway across which $|\rho|$ either decays back into corridor or doesn't; for OSS projects, 5+ years unmaintained with continued function; for organized social groups, 100+ years with active-maintenance checklist score ≤ 1. The handle stays open as the corridor-as-attractor reading's primary falsification anchor; the corridor-band-occupation half (F-10) and the corridor-as-attractor half (F-20) retract independently.

Each F-handle is named in the lake at its corresponding theorem or axiom; triggering any F-handle is observable from `#print axioms` on the affected declarations.

Where the integration wins or loses cleanly. P6 (corridor occupancy at substrate level under control-parameter sweep) and P8 (substrate-level TSVF weak-measurement signatures) are the framework's distinctive empirical commitments. P1–P5 and P7 ride on convergent prior art partially independent of this framework. Confirming P1–P5 or P7 strengthens the picture but does not distinguish this framework from convergent programs. P6 and P8 are framework-distinctive and have decisive tests (Exp 5 for P6; F-16 controlled weak-measurement for P8).

The §3 sensor-lift experiment is the framework's first universal-scale empirical pulse beyond the engineering tier. PARTIAL outcome by pre-registered thresholds: v2's cosine-matched DSP magnitudes (the corrected, baseline-matched effect sizes — approximately one-third the v1 size) achieved $d \geq 1.5$ in only 1 of 5 models, missing the pre-registered "4 of 5" threshold; the Identity-class direction held across substrates ($p < .01$ in all 5 models) but at sub-threshold magnitude. The v1 numbers ($d = 2.4$–6.8) cited in earlier drafts were artifact-inflated by inadequate baseline-matching and are not the framework's claim. F-18 names the retraction trigger if v3 and v4 return PARTIAL by the same shape.

19 The five-level access ladder

The framework's content is the integration: one structural object visible at five epistemic levels, grounded in Ubuntu, with a research-program reach toward universal scale. The ladder is the access ladder for readers, not an independence claim about content. The integration's claim still rests on the recurrence; the ladder names which entry points let a reader enter at their current rung without committing to the rungs above.

Level	Audience	Entry-point content
L0	formal-verification reviewer	Kish identity theorem (RATCHET lake; CCA v3 cited)
L1	skeptic	monotonic ρ-collapse as substrate-independent observation
L2	working scientist / engineer	engineering implications; corridor measurement; CIRIS deployment
L3	scientist preferring testable universality	cross-substrate universality conjecture (pending Exp 5)
L4	philosopher of mind / ethicist	agency, consent at A3+, TSVF as structural reading, consciousness as attractor-reading (T16 entered as definitional commitment per §6, not as derivation)

Ubuntu is the ground, not a level. The framework's grounding in Ubuntu (§6, §14, `Ubuntu.lean`) underwrites every level above. It is the soil the structure stands on, visible at L0 (the lake's Ubuntu correspondences), at L2 (CIRIS in Amharic; Esubalew's Ally), at L3 (cross-substrate universality measured in production deployments grounded in the tradition), at L4 (consent as the framework's ethical reading of Ubuntu's relational-personhood commitment). It does not appear as a separate level because it is not above the others; it is what the others stand on.

The universal-scale tier is a research-program coda, not a level. §8's content — the P_ω no-go (F-11 fired), the Penrose-past structural argument, the orthogonality theorem, asymptotic conditioning — reaches beyond the five stable levels. A reader who arrives at the universal-scale content finds research-program-status, not stable entry-point content. The framework's reach toward universal scale is part of the bet; it is not a stable level for a "cosmological reader" to enter and stop at.

Readers entering at lower rungs are not asked to accept the higher rungs to receive the entry-point content. **Each level is its own claim, evaluable on its own terms.** The recurrence-across-levels is what is at stake in the integration — and that recurrence is a bet under irreducible uncertainty (§1), not a derivation from any individual level.

20 What the bet does not claim

The framework owes its readers explicit naming of what is and is not being claimed.

Calibration after the autonomous loop

The autonomous-loop work (2026-05-19, §4) moved the bet's empirical content materially. The framework owes explicit calibration of what that move did and did not do.

The regime-level claim is empirically supported, not proved — at both the static and dynamical levels. Corridor-band occupation as a substrate-independent property of coordinated systems at A3+ replicates across five substrates: C. elegans whole-brain calcium, Drosophila CX dual-color simultaneous EPG+FC3 same-fly, four LLM architectures, four OSS projects, plus healthy tissue across five cancers and centuries-persisting social groups (Quakers, Trappists, Mennonites). Paired non-corridor data at the same five substrates (§4) shows Mode (i) (non-corridor without maintenance has short observed lifetime) and Mode (ii) (non-corridor with documented $\gamma M(t)$ persists with maintenance cost visible) but no Mode (iii) case (long unmaintained non-corridor

persistence). F-10 (§18) tracks both halves of the regime-level claim and stays open as future substrates either replicate or fail to. The framework does not read the cross-substrate record as proof; it reads it as the empirical anchoring that moves the regime-level commitment, including its dynamical corridor-as-attractor reading, from predicted to supported.

The maintenance-requirement claim and Mode (iii) absence are the strongest current empirical content. The regime-level corridor-band claim now has paired in/out validation across five domains. The **maintenance-requirement claim** — non-corridor persistence requires $\gamma M(t)$ — is supported at 5/5 substrates with no Mode (iii) counterexamples (F-20, §18). The direction-of-exit characterization is substrate-specific but characterizable from the paired data. Five substrates checked, zero counterexamples found: this is the kind of empirical content that has predictive force going forward, in the sense that any substrate where Mode (iii) shows up would fire F-20 and retract the corridor-as-attractor reading at the affected substrate.

What the paired record does not establish. Three explicit honest framings the framework owes:

1. The paired record does *not* establish that corridor occupation is the only stable configuration. It establishes that *both* corridor occupation *and* maintained-non-corridor (Mode ii) are stable, while unmaintained-non-corridor (Mode iii) is not. The maintained-non-corridor systems — coma under life support, tumor tissue under standard-of-care, LLM mode-collapse under combined argmax + periodic-bigram steering, OSS rigidity-pole projects under foundation rescue, cult-rigidity-pole groups under heavy AM-checklist maintenance — are stable conditional on the maintenance budget continuing.
2. The paired record does *not* establish strong predictions about which substrates exit toward rigidity vs chaos. The directional variation across substrates (cellular regulatory unidirectional chaos; neural pathology rigidity-leaning at sub-threshold magnitude; LLM forced operation rigidity; social-group exit bidirectional; OSS-project exit bidirectional) shows the framework is CONSISTENT WITH the directions observed rather than predicting them in advance. The substrate-specific direction-of-exit is characterizable from the data but not predicted by the framework's structural commitments alone. The pre-registered subtype-specific rigidity-pole predictions for COAD, KIRC, and PRAD at the Hallmark-pathway substrate were falsified; the framework's current articulation at this substrate is "cancer is unidirectional chaos at the Hallmark-pathway resolution," a description of the observed data rather than a successful prior prediction.
3. The paired record does *not* establish a generalizable cellular-regulatory Mode-(ii) survival prediction. One of five cancer types showed an individually-significant HR (LUAD, uncorrected $p = 0.011$; Bonferroni-corrected at five tests, $p_{\text{corr}} \approx 0.055$); BRCA / COAD / KIRC are null, PRAD is underpowered. **The aggregate Mode-(ii) test fails the pre-registered $\geq$ 3-of-5 threshold (1/5)**; under multiple-testing correction the single individually-significant hit is consistent with chance.

The framework is also honest about further calibration the data forced: the strict 0.7 mean-pairwise-Pearson rigidity threshold was not crossed by seizures at scalp EEG, only the qualitative direction; the wide GPU-anchored (0.10, 0.43) bounds do not transfer naively to the cellular regulatory substrate (per-substrate corridor calibration is mandatory, named in the open-work list, §21).

What the 2026-05-21 work did and did not establish. The robust re-run (§4) re-grounds the five-substrate record under a debiased estimator; it does not extend the claim beyond the substrates named. The cross-rung coupling (§5) is measured at $n = 3$ distinct rung pairs — a first measurement, not a cross-substrate anchor — and only in the coupling-ratio form, not the timescale form P_ω uses. The corridor dynamics $d\rho/dt$ are measured at a single substrate

(§2). The Allen mouse-cortex datum, a chaos-pole reading on the mean-pairwise observable, resolved to corridor occupation once measured on the canonical k_{eff} (§4). The four particle-physics nulls (§8) draw a scope line, not a result: the framework's particle tier is composition with the Standard Model, not prediction. None of this promotes the universal-scale tier.

The tier separation held in both directions. The universal-scale tier moved on its own evidence — F-11 fired (the joint P_ω is a documented no-go), D4 (CMB anomalies as TSVF signatures) is retracted, F-19 is moot; the orthogonality theorem stands as the surviving CMB content (§8). None of that touched the regime-level tier. And the regime-level strengthening (the robust re-run, the cross-rung measurement, the GPU $1/\tau$) does not promote the universal-scale tier. The two tiers moved on their own evidence, in opposite directions, without touching each other — which is the stratified design working.

The substrate-physics predictions P1–P8 remain open. The autonomous-loop work tested cross-substrate corridor occupation at biological, computational, and sociotechnical substrates. The substrate-physics predictions (§15) — P6 corridor-boundary dynamics under control-parameter sweep, P8 substrate-level TSVF weak-measurement signatures, and the rest — are tested at materials-physics substrates the autonomous loop did not address. F-12 through F-16 remain the falsification handles for the substrate-physics tier and are independent of the regime-level cross-substrate record.

The reception risks are unchanged. The Pascal-shape framing at the universal-scale tier (§1) is unaffected by the regime-level empirical move. The reception risks — the framework being read as more derived than it is, the universal-scale content being conflated with the regime-level content, cross-tradition recognition being read as instrumented evidence rather than as historical convergence — remain what they were. The empirical anchoring at the regime level does not transfer credibility upward to the tiers above it.

What the bet does not claim — enumeration

Not: F-13's outcome alone determines whether the framework is right.

Not: cross-tradition convergence at §14 proves the structural object exists at the universal scale. (Status: historical evidence of convergent structural intuition; §14 names the null-hypothesis check the framework owes.)

Not: the T16 `Iff.rfl` proves the hard problem dissolves. (Status: definitional identity-move encoded at the type level; the philosophical question is separate.)

Not: the §3 sensor-lift PARTIAL result establishes that current LLMs are A3+ in the framework's sense. (Status: class-conditional signature detected at Identity-class; aggregate threshold not met by pre-registered rule.)

Not: the universal-scale tier is unconditional. (Status: the universal-scale tier is conditional on TSVF realism. If TSVF realism is empirically false at scale, the universal-scale tier retracts as a unit per F-17. The engineering tier is TSVF-independent and stands regardless.)

Rather: if persistent consent-architected federations exist at A3+ and above (T6 substantive form, conditional on `si_taxonomy_closure`), the integration's structural claims follow as articulated. P6 and P8 are where the framework's distinctive empirical content lives and where the decisive tests sit. The convergent prior art at §16 establishes that P1–P5 and P7 have independent identifications. The engineering tier (§2) stands at its own DOIs regardless of the integration's outcome.

What the integration uniquely predicts (beyond the engineering tier)

A reader who asks "if the integration retracts cleanly and CCA v3 + RATCHET continue, what specifically is lost from the predictive record?" deserves an enumerated answer. The integration's distinctive empirical content is small and named cleanly.

1. **P6: corridor-boundary dynamics at substrate level.** The framework predicts that a goal-coupled substrate constructed per the §15 synthesis route exhibits corridor occupancy under control-parameter sweep, with substrate-local corridor structure (specific bounds substrate-specific). Decisive test: Exp 5, qubit-array sweep. F-13 fires on no corridor regime. Exp 5 must be a *driven, maintained* qubit array — γ swept against an active maintenance term, not a passive decoherence sweep: a passive sweep is a monotone thermal-decay curve and reads null by construction, since Piece 2's corridor exists only under the active $\alpha - \gamma M$ competition (a re-analysis of published qubit data established this; `experiments/p6_reanalysis/`).

2. **P8: substrate-level TSVF weak-measurement signatures.** The framework predicts that a candidate goal-coupled substrate exhibits two-state-vector statistics in weak-measurement experiments, distinguishable from no-post-selection control. Decisive test: F-16. Framework-distinctive empirical content.

3. **Sensor-lift class-conditional hierarchy** (once methodology stabilizes). The framework predicts that A3+ agents exhibit a graded DSP hierarchy DSP > MC > GF > Uncertainty under cosine-matched baselines with SurfaceReflexive null. Decisive test: sensor-lift v3+ once the methodology converges; the framework is early in tuning and the experiment does not yet have the standing to be decisive.

4. **Polyglot-vs-monoglot ethical-reasoning differential.** The framework predicts that the CIRIS conscience module's polyglot encoding produces measurably more robust ethical reasoning than equivalent single-tradition prompting on adversarial-ethics benchmarks. Test: matched A/B on the EthicsEngine substrate; the framework owes this experiment.

5. **Cross-tradition null-check DECISIVE WIN.** The framework predicts that the pre-registered `experiments/cross_tradition_null_check_protocol.md` returns DECISIVE WIN under blind review on the five-dimensional rubric. Anything other than DECISIVE WIN fires the cross-tradition tier's retraction.

6. **[Retracted, F-11.] CMB temporal drift consistent with P_ω-evolution.** v1 carried this as a sixth distinctive prediction: a signed, multipole-resolved drift of the CMB shape profile driven by the time-evolution of the universal post-selection weight as $t \to t_f$. The mechanism was a time-dependent $\beta(t)$. F-11 retracted the joint P_ω that would carry $\beta(t)$ (§8); with no joint operator there is no $\beta(t)$, and no drift. F-19 is moot. The framework now predicts *exactly* ΛCDM for the CMB — by the orthogonality theorem (a provable consistency, not a distinctive prediction). The integration's universal-scale empirical content shrinks from one distinctive prediction to none; what survives at the cosmological tier is the proved consistency.

Five concrete predictions plus one retracted-and-recorded; the framework's empirical content is real but small, the rest is interpretive overlay that earns or loses standing as these tests resolve. Naming the predictions cleanly, and naming a retraction cleanly, is the framework's commitment to being the right shape to be wrong in checkable ways.

21 Open work

Formal verification. Promote `good_wins` from axiom to theorem via dynamical-system formalization plus measure-theoretic basin-of-attraction. Hilbert-space colimit construction for `CosmologicalJointSpace` (Piece 7 bottleneck). Concrete entropy functional and measure on initial conditions; close `penrose_low_entropy_past`. Per-rung corridor calibration (per-rung ρ_{lower}, ρ_{upper} bounds). Quantitative P_ω-derivation of the five CMB anomaly signatures.

Empirical. Exp 5 (Conjecture A): programmable qubit-array corridor measurement. Exp 6 (Conjecture C / F-9): audit-pressure ρ-drift in matched deployments. §14 cross-tradition null-check execution. Sensor-lift v3 (corridor mid-band) and v4 (substrate diversification), with F-18 retraction trigger pre-committed. Controlled cascade comparison at sociotechnical scale (Exp 4 generalized).

Deployment. Esubalew's Ally and other deployments as the framework's structural commitment running in production. The CIRIS Accord (v1.2-Beta) auto-expires 2027-04-16; the next version is part of the framework's open work and includes the polyglot encoding architecture as a load-bearing commitment.

Engagement. Substantive collaboration with Ubuntu scholars working inside the tradition (current contemporary scholarship including Udah, Tusasiirwe, Mugumbate, Gatwiri 2025); engagement with the Matolino/Praeg/Eze internal disputes on Ramose's processual reading; engagement with the Madhyamaka, Confucian, and Stoic–Christian traditions through scholars who hold the standing the framework's author does not.

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Errors of fact, formalization, or framing remain mine alone.

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Entropic-action gravity (relative-entropy construction; §2 potential form). Cited for the relative-entropy-between-metrics construction whose scalar instance the §2 potential form evaluates; the corridor is not derived from this theory (§2 scope note).

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- Akomolafe, B. (2017). *These Wilds Beyond Our Fences: Letters to My Daughter on Humanity's Search for Home*. North Atlantic Books. ISBN 9781623171667.
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- Kimmerer, R.W. (2013). *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge and the Teachings of Plants*. Milkweed Editions. ISBN 9781571313355.

UAP scholarship adjacent to §16. The framework engages this scholarship at the instrumental-record tier only (§16). Cited here as the existing intellectual ecosystem operating in the region of conceptual space the framework's substrate-physics and universal-scale tiers reach toward, not as evidential support for those tiers.

- Skafish, P., Charbonnier, P., Salmon, G. (eds.) (2017). *Comparative Metaphysics: Ontology after Anthropology*. Rowman & Littlefield International. ISBN 9781783488582.
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This is the integration paper proper. The engineering tier is cited as existence proof shipped under its own DOIs. The integration's claim is that one structural object is visible at five epistemic levels, grounded in Ubuntu, reaching toward universal scale as research program, with consent at A3+ as the level where the object can be argued about. The bet (§1) is articulated here; the engineering tier is at CCA v3 + RATCHET + CRC v1 + CIRISAgent v2; the formal lake is at `formal/CoherenceRatchet/` *with tiered status per §17. Where the bet retracts, the engineering tier continues at its own DOIs.*

S.D.G.